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Greater China Technology Hardware | Asia Pacific

# ABF Substrates: Pricing Upcycle Is Accelerating

The substrate pricing cycle is proving stronger and earlier than expected. As pricing accelerates and margins continue to expand, we see upside to earnings. We raise estimates across our coverage and continue to recommend owning UMTC (preferred play), NYPCB, and ZDT.

**Stronger-than-expected 2Q profitability supports another round of estimate revisions:** Unimicron and NYPCB both reported preliminary 2Q earnings that exceeded our expectations. Unimicron's preliminary April earnings came in ahead of our estimates and Street expectations ([link](#)), while NYPCB's preliminary April ([link](#)) and May ([link](#)) results also outperformed both. We attribute the upside primarily to higher utilization rates, a more favorable product mix, and stronger-than-expected pricing. These developments provide the basis for raising our earnings estimates for Unimicron, NYPCB, and Zhen Ding.

**Substrate pricing is improving faster than expected; further upside likely in 2H26:** Our latest supply chain checks indicate that pricing momentum for both BT and ABF substrates continues to strengthen. Specifically, we estimate NYPCB's BT substrate pricing increased by ~20–30% during 2Q, while ABF substrate pricing rose by roughly 10%. We expect additional pricing increases in 2H26 as supply constraints continue to tighten. We believe this pricing improvement was the primary driver of NYPCB's stronger April and May profitability, and, assuming pricing continues to improve, we expect gross margins to expand further in 3Q. Although we believe these pricing dynamics should benefit the broader substrate industry, NYPCB has historically been among the most aggressive in implementing price increases and may therefore continue to outperform peers on pricing.

**Are BT substrate suppliers facing pricing pressure from memory makers?** Based on our checks, we do not believe this is the case, at least for Taiwanese BT substrate suppliers. On the contrary, lead times continue to extend, while some suppliers, including Unimicron, appear to be reducing capacity, which further tightens supply.

**We reiterate Overweight on Unimicron, NYPCB, and Zhen Ding:** Reflecting stronger earnings trajectories and improved pricing outlooks, we raise our EPS estimates for all three companies while reiterating our positive investment stances. Based on upside to our revised price targets, our order of preference is **Unimicron (NT\$1,465 PT; 25x CY28 P/E; 50% upside)**, followed by **NYPCB (NT\$1,550 PT; 25x CY28 P/E; 35% upside)**, and **Zhen Ding (NT\$690 PT; 20x CY28 P/E; 15% upside)**.

**Potential risks:** 1) Weaker demand for PCs, general servers, and AI servers; 2) meaningful capacity expansion plans; 3) T-glass being more constrained than we expect; 4) CoWoP displacing ABF substrates.

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Exhibit 1 : What's changed?

| Company   | Ticker  | Estimate |      | 2Q      |         | Last<br>close | EPS<br>(US) | EPS change (%) |     |     |      | P20  |      |     |
|-----------|---------|----------|------|---------|---------|---------------|-------------|----------------|-----|-----|------|------|------|-----|
|           |         | Rev      | Cost | Rev     | Cost    |               |             | 2Q             | 25x | 25x | 25x  | 25x  | 25x  | 25x |
| Unimicron | 3037.TW | CW       | CW   | 1,400.0 | 1,200.0 | 970.0         | 50%         | 20%            | 20% | 14% | 0.41 | 0.35 | 0.34 |     |
| Nypcb     | 3046.TW | CW       | CW   | 1,200.0 | 1,270.0 | 1140.0        | 20%         | 20%            | 20% | 22% | 0.24 | 0.42 | 0.37 |     |
| Zhen Ding | 4058.TW | CW       | CW   | 690.0   | 660.0   | 550.0         | 10%         | 4%             | 4%  | 4%  | 0.30 | 0.40 | 0.34 |     |

Source: Bloomberg, Morgan Stanley Research (e) estimates. Pricing as of the close on July 2, 2026.

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Preview to earnings

| Focus KPI                                                                                                                                                       | Focus KPI Surprise         | Likely impact to consensus EPS* |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------|
| <b>Unimicron</b> 3037.TW                                                                                                                                        |                            |                                 |
| 2Q GM and 3Q revenue outlook                                                                                                                                    | ↑ Likely upside surprise   | ↑ Modest revision higher        |
| • We raise our 2Q GM estimate to 22.1% (vs 19.5% previously) and are now 80bps above the Street.                                                                |                            |                                 |
| <hr/>                                                                                                                                                           |                            |                                 |
| <b>Nan Ya PCB</b> 8046.TW                                                                                                                                       |                            |                                 |
| 2Q GM and 3Q revenue outlook                                                                                                                                    | ↑ Likely upside surprise   | ↑ Modest revision higher        |
| • Our 2Q GM forecast is 60bps above the Street. We raise our 2Q EPS estimate by 31% to NT\$3.35 and are now 7% above the Street.                                |                            |                                 |
| <hr/>                                                                                                                                                           |                            |                                 |
| <b>Zhen Ding</b> 4958.TW                                                                                                                                        |                            |                                 |
| 2Q GM and 3Q margin outlook                                                                                                                                     | ↓ Likely downside surprise | — Largely unchanged             |
| • Our 2Q GM forecast is 70bps above the Street. We raise 2Q EPS estimate by 6% to NT\$2.09 but are still 5% below the Street, reflecting higher opex and taxes. |                            |                                 |

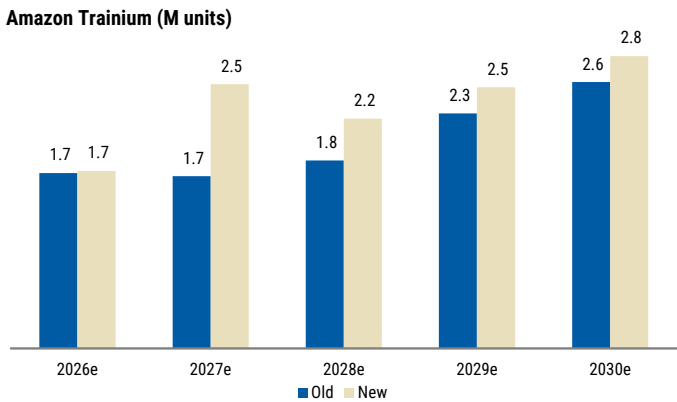
\*Likely impact to consensus EPS is for the next 12 months  
Source: Company data, Morgan Stanley Research

# Changes to Our Assumptions

## Stronger ASIC demand from hyperscalers

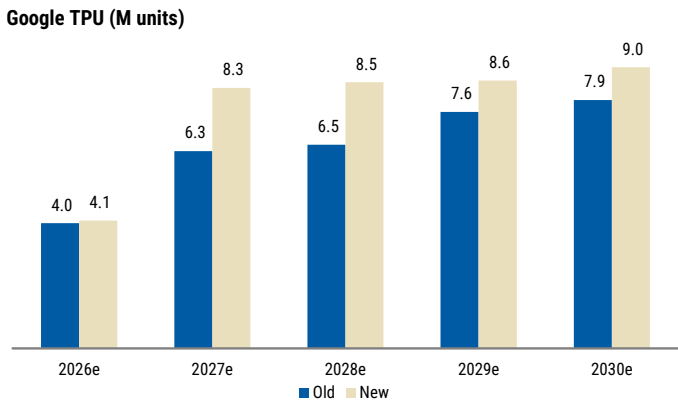
We have updated our ASIC demand forecasts to reflect the latest estimates from our Semiconductors team. As shown in the exhibits below, our revised assumptions point to a steeper production ramp for Amazon's Trainium processors over 2027–30, as well as stronger demand for Google's TPU portfolio over the next five years. Given that these ASICs are characterized by large die sizes and high layer-count package substrates, the upward revisions imply incrementally stronger demand for ABF substrates, further reinforcing our constructive outlook for the substrate suppliers.

Exhibit 2: Changes to Amazon Trainium estimates



Source: Morgan Stanley Research (e) estimates.

Exhibit 3: Changes to Google TPU estimates



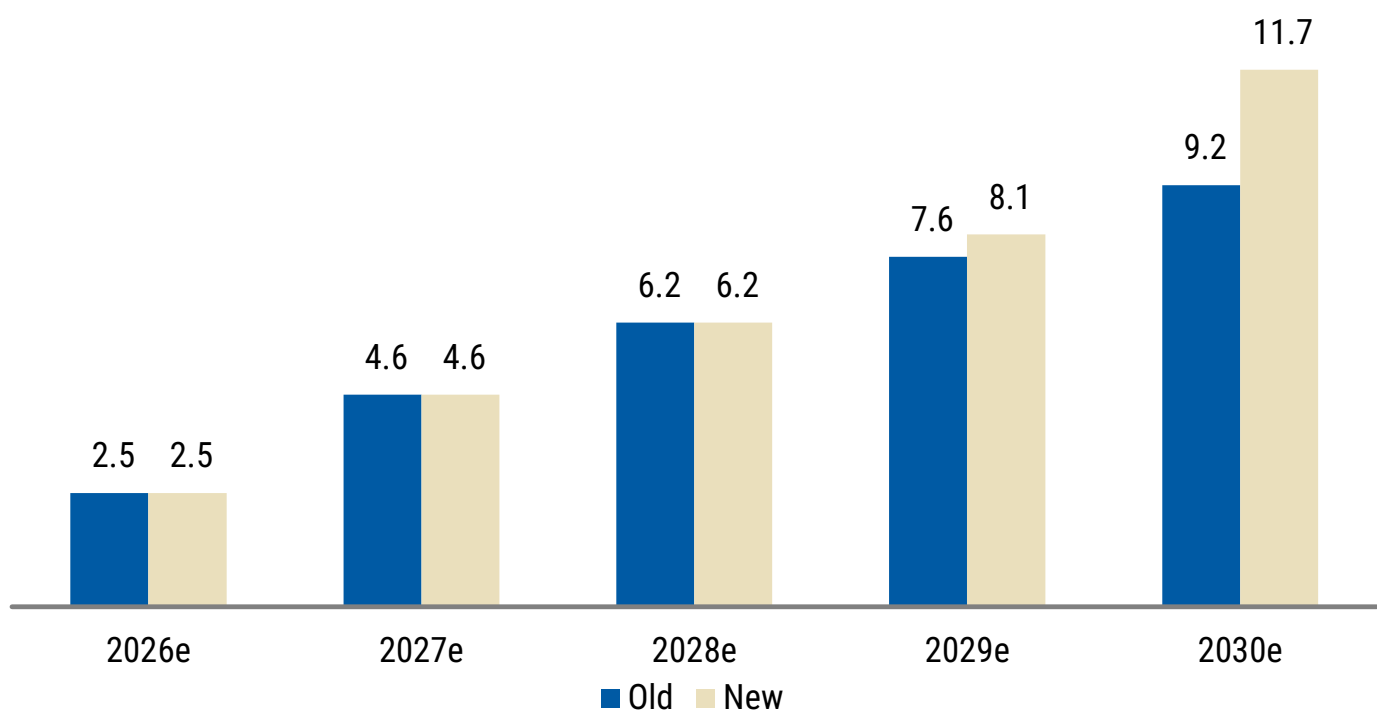
Source: Morgan Stanley Research (e) estimates.

## Other ASIC chips

For "Other" ASIC chips, we also increase our estimates. Included in this section are ASIC chips from Microsoft, other US companies, and China AI chips, such as those from Huawei, Cambricon, Hygon, T-Head, etc. For China AI chips, we included the latest estimates, also from our semiconductor team, which was published in [this report](#). The latest estimate for China AI chips TAM by 2030 is 36% higher than the prior assumption, driving total "Other ASIC chips" category to reach ~11.7M units by 2030, up from ~9.2M units previously.

Exhibit 4: Seeing increased demand from China AI chips

"Other ASIC" chip, includes China AI chips (M units)

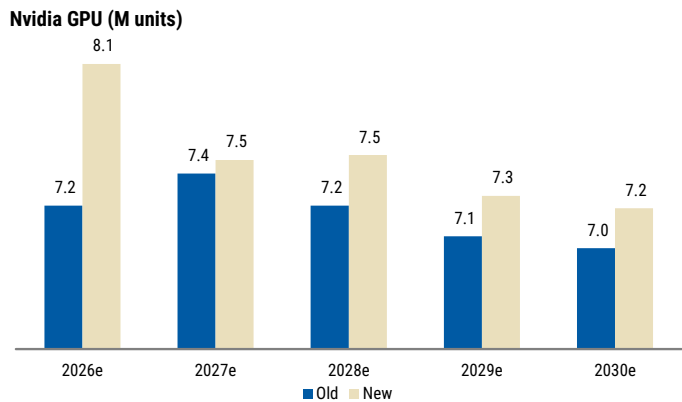


Source: Morgan Stanley Research (e) estimates.

Changes to our GPU forecasts

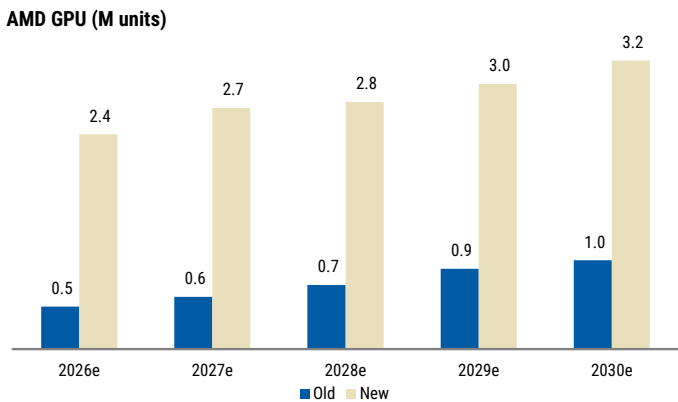
We also update our demand forecasts for Nvidia and AMD GPUs, with our revised estimates generally pointing to stronger demand over CY26–30 compared with our previous assumptions. In addition, we now explicitly incorporate forecasts for Nvidia's server CPU portfolio (including Grace, Vera, and future platforms) into our demand model. Together, these revisions result in a higher overall substrate demand outlook relative to our prior forecasts, reflecting both stronger AI accelerator demand and the growing contribution from Nvidia's server CPU roadmap.

Exhibit 5: Changes to our Nvidia GPU estimates



Source: Morgan Stanley Research (e) estimates.

Exhibit 6: Changes to our AMD GPU estimates

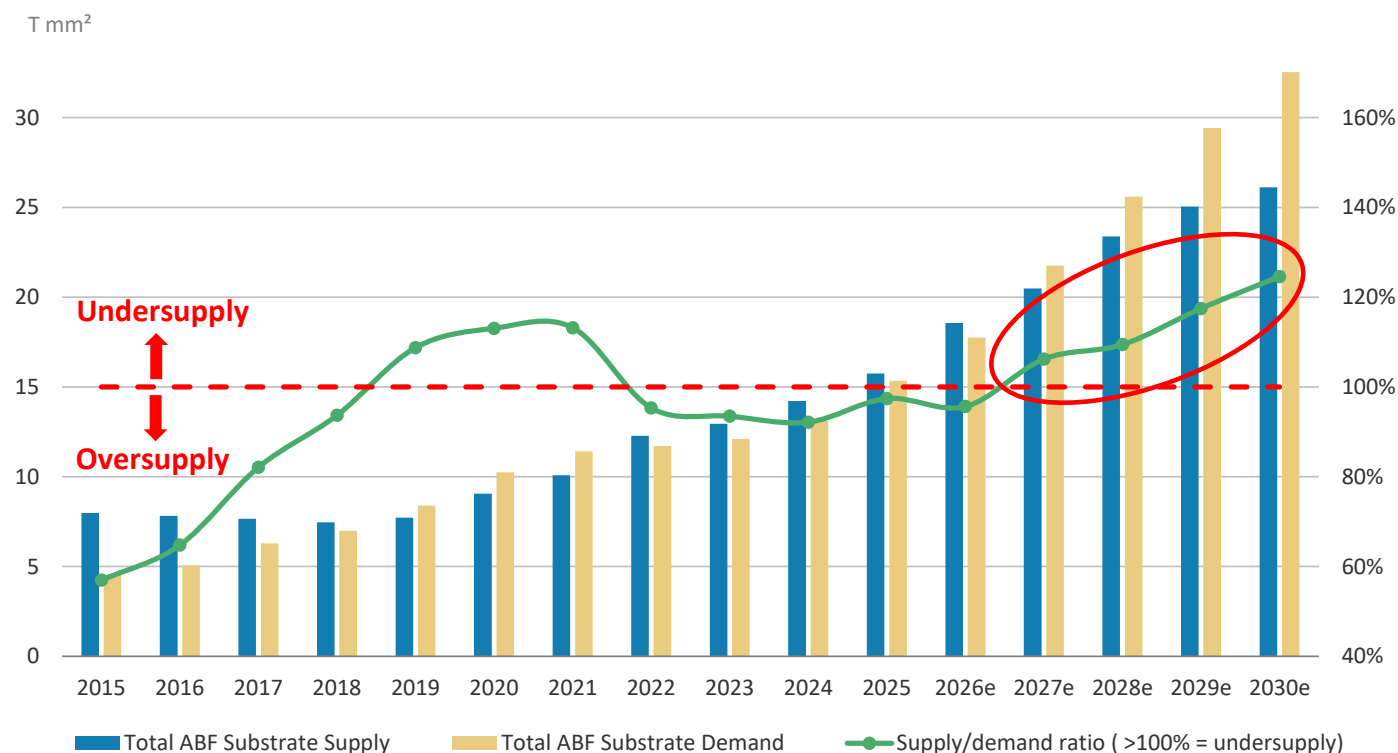


Source: Morgan Stanley Research (e) estimates.

**We also baked in recent capacity expansion announcement from AT&S in our model in this report:** AT&S (ATSV.VI, not covered) announced that it signed key terms with AMD and another major tech customer to add AI/HPC IC substrate capacity in Kulim, Malaysia. The company said the €1.5-2bn investment is fully supported and financed by long-term customer commitments, pending final execution.

**We now expect the ABF supply/demand imbalance to widen by 2030:** After baking in the latest assumptions and estimates, our bottom-up ABF substrate S/D model points to 25% undersupply by 2030 (vs. ~22% previously). A detailed breakdown of the changes in assumptions that resulted in this shortage widening can be found in the next section. But increased undersupply gives us more confidence about pricing and margin tailwinds that we expect to see for Unimicron, NYPCB, and ZDT.

**Exhibit 7:** Expecting ABF substrate under-supply gap to widen by 2030 vs. our prior estimates



Source: Morgan Stanley Research (e) estimates.

**Exhibit 8:** Increased demand assumptions cause CY29-30 under-supply gap to widen vs. our prior estimates

| ABF SD | 2026e | 2027e  | 2028e  | 2029e  | 2030e  |
|--------|-------|--------|--------|--------|--------|
| Old    | 93.3% | 102.2% | 105.8% | 114.9% | 122.3% |
| New    | 95.6% | 106.2% | 109.4% | 117.5% | 124.6% |

Source: Morgan Stanley Research (e) estimates.

## What does this new S/D dynamic mean for pricing and margins?

We estimate cost inflation will drive ABF substrate pricing to grow 20-25% y/y in CY26, and 25%+ y/y in CY27, mainly driven by an imbalance in ABF substrate S/D. Because our view is that the imbalance will continue to widen in CY28 and thereafter, we are also

assuming pricing will continue to move higher in CY28, in the range of 25-40% y/y, depending on the customer. For Unimicron, which has higher exposure to larger customers, we model 20%+ pricing growth y/y for CY28, but for NYPCB, we estimate pricing growth will accelerate in CY28 to 40%+ y/y growth, reflecting a more severe ABF substrate supply shortage.

**Any new capacity expansion plans will need at least two years to bring online:** As a result, any new meaningful capacity expansion plans that we hear from hereon out will affect S/D dynamics only for CY29 onward. So for the next two years, our focus will be on changes to end-demand when analyzing this space.

# Earnings Calendar and Comps Table

**Exhibit 9:** Earnings Calendar

| Company    | C2Q26 Result Date | Earnings Call |
|------------|-------------------|---------------|
| SEMC       | Late July         | Late July     |
| Ibiden     | August 4          | August 5      |
| AT&S       | August 4          | August 4      |
| Unimicron  | July 28           | July 29       |
| Nan Ya PCB | August 6 or 7     | No            |
| Zhen Ding  | By mid-August     | TBC           |

Source: Company data, Morgan Stanley Research.

## 2Q Preview: MSe vs. Consensus

### Unimicron

Our estimates are broadly in line with consensus at the revenue level but differed on the margin profile. We forecast net sales of NT\$42.5bn, essentially in line with consensus, while expecting a stronger gross margin of 22.1%, ~80bps above the Street estimate of 21.4%, reflecting a more favorable product mix and better pricing environment. However, we are also more conservative on operating expenses, resulting in an operating margin of 12.1%, modestly below consensus (12.3%). The key difference vs. the Street is below the operating line, where we assume materially lower non-operating income (NT\$491mn vs. NT\$980mn consensus), leading to pre-tax income and net income estimates that are 9% and 17% below consensus, respectively. Consequently, our EPS forecast of NT\$2.50 is ~17% below the Street estimate of NT\$2.99, despite our more constructive view on gross profitability.

**Exhibit 10:** Unimicron 2Q Preview: We are 80bps above the Street on 2Q GM

| Year to Dec. 31 (NT\$ mn) | Actual<br>2Q26E | Q/Q         | Y/Y           | Consensus<br>2Q26E | Diff. %     |
|---------------------------|-----------------|-------------|---------------|--------------------|-------------|
| <b>P&amp;L Summary</b>    |                 |             |               |                    |             |
| <b>Net sales</b>          | <b>42,509</b>   | <b>14%</b>  | <b>31%</b>    | <b>42,358</b>      | <b>0%</b>   |
| COGS                      | (33,108)        |             |               | (33,311)           |             |
| <b>Gross profit</b>       | <b>9,402</b>    | <b>40%</b>  | <b>121%</b>   | <b>9,047</b>       | <b>4%</b>   |
| Operating expenses        | (4,272)         |             |               | (3,855)            |             |
| <b>Operating income</b>   | <b>5,130</b>    | <b>86%</b>  | <b>240%</b>   | <b>5,192</b>       | <b>-1%</b>  |
| Non-operating income      | 491             |             |               | 980                |             |
| <b>Pre-tax income</b>     | <b>5,620</b>    | <b>-11%</b> | <b>1475%</b>  | <b>6,172</b>       | <b>-9%</b>  |
| <b>Net income</b>         | <b>3,877</b>    | <b>-23%</b> | <b>12996%</b> | <b>4,648</b>       | <b>-17%</b> |
| <b>EPS (NT\$)</b>         | <b>2.50</b>     |             |               | <b>2.99</b>        |             |
| <b>Margins</b>            |                 |             |               |                    |             |
|                           |                 | <b>ppt</b>  | <b>ppt</b>    |                    | <b>ppt</b>  |
| Gross margin              | 22.1%           | 4.2%        | 9.0%          | 21.4%              | 0.8%        |
| Operating margin          | 12.1%           | 4.7%        | 7.4%          | 12.3%              | -0.2%       |
| Pretax margin             | 13.2%           | -3.6%       | 12.1%         | 14.6%              | -1.4%       |
| Net margin                | 9.1%            | -4.3%       | 9.0%          | 11.0%              | -1.9%       |

Source: Company data, Visible Alpha, Morgan Stanley Research estimates.

## NYP CB

Our estimates are modestly ahead of consensus, driven primarily by stronger operating profitability. Although our revenue forecast is broadly in line with consensus (NT\$13.575bn vs. NT\$13.64bn consensus), we are more constructive on margins, particularly at the operating level. Our operating margin estimate of 18.2% is 70bps above consensus (17.4%), reflecting stronger gross margin (21.45% vs. 20.8% consensus) and lower-than-expected operating expenses. As a result, our operating profit forecast exceeds consensus by approximately 42%, which, when combined with slightly lower taxes, flows through to EPS of NT\$3.35, around 7% above the Street estimate of NT\$3.13. Overall, we assume earnings will come in better-than-expected driven more by better profitability than by top-line performance.

**Exhibit 11:** NYP CB 2Q Preview: We are 75% above the Street on 2Q EPS

| NT\$m                   | 2Q26E Results |            |             | Cons          | Variance   |
|-------------------------|---------------|------------|-------------|---------------|------------|
|                         | Actual        | QoQ        | YoY         |               |            |
| <b>Net sales</b>        | <b>13,575</b> | <b>21%</b> | <b>42%</b>  | <b>13,636</b> | <b>0%</b>  |
| COGS                    | -10,666       | NM         | NM          | -10,795       | -1%        |
| <b>Gross profit</b>     | <b>2,909</b>  | <b>64%</b> | <b>278%</b> | <b>2,841</b>  | <b>2%</b>  |
| Operating expenses      | -445          | NM         | NM          | -467          | -5%        |
| <b>Operating income</b> | <b>2,464</b>  | <b>82%</b> | <b>573%</b> | <b>2,374</b>  | <b>4%</b>  |
| Non-operating income    | 176           | -28%       | NM          | 175           | 1%         |
| <b>Pre-tax income</b>   | <b>2,640</b>  | <b>65%</b> | <b>NM</b>   | <b>2,548</b>  | <b>4%</b>  |
| Income tax              | -475          | NM         | NM          | -523          | -9%        |
| <b>Net income</b>       | <b>2,165</b>  | <b>65%</b> | <b>NM</b>   | <b>2,025</b>  | <b>7%</b>  |
| <b>EPS (NT\$)</b>       | <b>3.35</b>   | <b>65%</b> | <b>NM</b>   | <b>3.13</b>   | <b>7%</b>  |
| <b>Margins (%)</b>      |               |            |             |               |            |
|                         |               | <i>ppt</i> | <i>ppt</i>  |               | <i>ppt</i> |
| Gross margin            | 21.4%         | 5.6        | 13.4        | 20.8%         | 0.6        |
| <b>Operating margin</b> | <b>18.2%</b>  | <b>6.1</b> | <b>14.3</b> | <b>17.4%</b>  | <b>0.7</b> |
| Pre-tax margin          | 19.4%         | 5.2        | 21.8        | 18.7%         | 0.8        |
| Net margin              | 15.9%         | 4.2        | 17.9        | 14.9%         | 1.1        |

Source: Company data, Visible Alpha, Morgan Stanley Research estimates.

## Zhen Ding

Our estimates are modestly above consensus at the operating level, although more conservative below the line. We forecast revenue of NT\$48.2bn, ~2% above consensus, reflecting stronger demand assumptions. We also expect a higher gross margin of 22.8%, 70bps above the Street estimate of 22.1%, resulting in gross profit 6% above consensus. Despite assuming slightly higher operating expenses, our operating income estimate remains 4% above consensus, supported by stronger gross profitability. Below the operating line, however, we are more conservative on non-operating income (NT\$314mn vs. NT\$352mn consensus), leading to net income and EPS forecasts ~5% below the Street. Overall, the key difference vs. consensus is that our model assumes stronger core operating performance but more modest non-operating contributions, offsetting the operating beat on the bottom line.



**Exhibit 12:** Zhen Ding 2Q Preview: We are 5% below the Street on 2Q EPS

| NT\$m                   | 2Q26E Results |            |             | Cons          | Var.       |
|-------------------------|---------------|------------|-------------|---------------|------------|
|                         | 2Q26E         | QoQ        | YoY         |               |            |
| <b>Net sales</b>        | <b>48,160</b> | <b>18%</b> | <b>26%</b>  | <b>47,010</b> | <b>2%</b>  |
| COGS                    | -37,187       | 17%        | 19%         | -36,616       | 2%         |
| <b>Gross profit</b>     | <b>10,972</b> | <b>25%</b> | <b>57%</b>  | <b>10,394</b> | <b>6%</b>  |
| Operating expenses      | -6,875        | 9%         | 50%         | -6,458        | 6%         |
| <b>Operating income</b> | <b>4,097</b>  | <b>64%</b> | <b>69%</b>  | <b>3,936</b>  | <b>4%</b>  |
| Non-operating income    | 314           | -404%      | -305%       | 352           | -11%       |
| <b>Pre-tax income</b>   | <b>4,411</b>  | <b>84%</b> | <b>94%</b>  | <b>4,288</b>  | <b>3%</b>  |
| <b>Net income</b>       | <b>2,242</b>  | <b>57%</b> | <b>271%</b> | <b>2,350</b>  | <b>-5%</b> |
| <b>EPS (NT\$)</b>       | <b>2.09</b>   | <b>57%</b> | <b>231%</b> | <b>2.20</b>   | <b>-5%</b> |

| Margins (%)      |       | ppt | ppt |       | ppt  |
|------------------|-------|-----|-----|-------|------|
| Gross margin     | 22.8% | 1.1 | 4.4 | 22.1% | 0.7  |
| Operating margin | 8.5%  | 2.4 | 2.2 | 8.4%  | 0.1  |
| Pre-tax margin   | 9.2%  | 3.3 | 3.2 | 9.1%  | 0.0  |
| Net margin       | 4.7%  | 1.2 | 3.1 | 5.0%  | -0.3 |

Source: Company data, Visible Alpha, Morgan Stanley Research estimates.

## MSe vs. Consensus

**Exhibit 13:** Unimicron: MSe vs. Consensus Estimates

| Unimicron  | MSe     |         |         | Consensus |         |         | Variance |      |      |
|------------|---------|---------|---------|-----------|---------|---------|----------|------|------|
|            | 2026    | 2027    | 2028    | 2026      | 2027    | 2028    | 2026     | 2027 | 2028 |
| Revenue    | 184,755 | 265,117 | 355,074 | 181,603   | 256,760 | 344,427 | 2%       | 3%   | 3%   |
| GM         | 23.1%   | 31.9%   | 38.3%   | 23.0%     | 32.0%   | 36.7%   | 0%       | 0%   | 2%   |
| OpM        | 13.6%   | 24.4%   | 31.8%   | 13.9%     | 24.2%   | 29.3%   | 0%       | 0%   | 2%   |
| Net income | 22,855  | 51,521  | 89,367  | 23,299    | 49,297  | 79,682  | -2%      | 5%   | 12%  |
| EPS        | 14.85   | 33.48   | 58.07   | 15.14     | 32.03   | 51.77   | -2%      | 5%   | 12%  |

Source: Visible Alpha, Morgan Stanley Research estimates.

**Exhibit 14:** NYPCB: MSe vs. Consensus Estimates

| NYPCB      | MSe    |        |         | Consensus |        |         | Variance |      |      |
|------------|--------|--------|---------|-----------|--------|---------|----------|------|------|
|            | 2026   | 2027   | 2028    | 2026      | 2027   | 2028    | 2026     | 2027 | 2028 |
| Revenue    | 56,808 | 78,966 | 111,555 | 58,647    | 90,842 | 125,480 | -3%      | -13% | -11% |
| GM         | 23.0%  | 33.8%  | 44.8%   | 23.1%     | 32.8%  | 39.3%   | 0%       | 1%   | 5%   |
| OpM        | 19.9%  | 31.2%  | 42.6%   | 19.7%     | 30.1%  | 36.9%   | 0%       | 1%   | 6%   |
| Net income | 10,108 | 21,337 | 40,621  | 9,795     | 23,137 | 38,756  | 3%       | -8%  | 5%   |
| EPS        | 15.64  | 33.02  | 62.87   | 15.16     | 35.81  | 59.98   | 3%       | -8%  | 5%   |

Source: Visible Alpha, Morgan Stanley Research estimates.

**Exhibit 15:** Zhen Ding: MSe vs. Consensus Estimates

| Zhen Ding  | MSe     |         |         | Consensus |         |         | Variance |      |      |
|------------|---------|---------|---------|-----------|---------|---------|----------|------|------|
|            | 2026    | 2027    | 2028    | 2026      | 2027    | 2028    | 2026     | 2027 | 2028 |
| Revenue    | 226,233 | 291,509 | 337,259 | 223,948   | 288,340 | 351,953 | 1%       | 1%   | -4%  |
| GM         | 24.5%   | 27.0%   | 29.1%   | 23.8%     | 25.7%   | 27.1%   | 1%       | 1%   | 2%   |
| OpM        | 12.0%   | 15.6%   | 18.6%   | 11.2%     | 14.5%   | 16.5%   | 1%       | 1%   | 2%   |
| Net income | 16,139  | 27,071  | 37,047  | 15,536    | 25,702  | 35,394  | 4%       | 5%   | 5%   |
| EPS        | 15.08   | 25.30   | 34.62   | 14.52     | 24.02   | 33.07   | 4%       | 5%   | 5%   |

Source: Visible Alpha, Morgan Stanley Research estimates.

**Exhibit 16:** ABF substrate peer comparison table

| ABF Substrate Ticker | Company   | Closing price (LC)<br>7/3/2025 | Rating | Price target | Market Cap<br>(US\$ mn) | EPS (LC)  |           |           | Current P/E (X) |      |      | PT PEG (X) |      |      | P/B (X) |      |     | P/S (X) |      |     | EV/EBITDA |      |      | ROA (%) |     |     | ROE (%) |     |     |
|----------------------|-----------|--------------------------------|--------|--------------|-------------------------|-----------|-----------|-----------|-----------------|------|------|------------|------|------|---------|------|-----|---------|------|-----|-----------|------|------|---------|-----|-----|---------|-----|-----|
|                      |           |                                |        |              |                         | 26E       | 27E       | 28E       | 26E             | 27E  | 28E  | 26E        | 27E  | 28E  | 26E     | 27E  | 28E | 26E     | 27E  | 28E | 26E       | 27E  | 28E  | 26E     | 27E | 28E | 26E     | 27E | 28E |
| 3037.TW              | Unimicron | 969                            | OW     | 1,465        | 45,728                  | 14.85     | 33.48     | 58.07     | 65.2            | 28.9 | 16.7 | 0.27       | 0.23 | 0.23 | 11.8    | 8.9  | 6.4 | 7.9     | 5.5  | 4.1 | 31.0      | 15.9 | 9.5  | 8%      | 15% | 20% | 20%     | 35% | 45% |
| 8046.TW              | NYPGB     | 1,185                          | OW     | 1,550        | 24,018                  | 15.64     | 33.02     | 62.87     | 75.7            | 35.9 | 18.8 | 0.18       | 0.32 | 0.21 | 14.0    | 11.0 | 8.0 | 13.5    | 9.7  | 6.9 | 43.1      | 24.2 | 13.3 | 15%     | 25% | 38% | 20%     | 34% | 49% |
| 4958.TW              | Zhen Ding | 613                            | OW     | 690          | 20,574                  | 15.08     | 25.30     | 34.62     | 40.6            | 24.2 | 17.7 | 0.34       | 0.36 | 0.48 | 3.6     | 3.2  | 2.9 | 2.9     | 2.3  | 1.9 | 17.6      | 11.7 | 8.7  | 6%      | 8%  | 11% | 9%      | 14% | 17% |
| 4062.T               | Ibiden    | 23,345                         | UW     | 13,000       | 40,463                  | 213.12    | 293.93    | 381.88    | 109.5           | 79.4 | 61.1 | -131.70    | 2.09 | 2.04 | 10.8    | 9.5  | 8.3 | 12.8    | 10.8 | 8.9 | 36.9      | 29.0 | 23.5 | 6%      | 8%  | 10% | 11%     | 14% | 16% |
| 009150.KS            | SEMCO     | 1,989,000                      | OW     | 2,560,000    | 97,578                  | 23,555.40 | 55,476.57 | 78,349.18 | 84.4            | 35.9 | 25.4 | 0.54       | 0.26 | 0.62 | 13.9    | 10.2 | 7.3 | 10.2    | 6.6  | 4.8 | 42.1      | 21.3 | 15.0 | 11%     | 22% | 25% | 18%     | 33% | 34% |
| ATS.AV               | AT&S      | 190                            | NC     | --           | 8,429                   | 4.21      | 7.05      | 12.86     | 45.0            | 26.9 | 14.7 | -0.04      | 0.40 | 0.18 | 6.5     | 5.5  | 3.6 | 3.1     | 2.5  | 2.0 | 6.3       | 6.3  | 6.3  | 3%      | 6%  | 9%  | 15%     | 20% | 25% |
| 3189.TW              | Kinsus    | 830                            | NC     | --           | 13,691                  | 10.23     | 22.27     | 42.34     | 81.2            | 37.3 | 19.6 | 0.41       | 0.32 | 0.22 | 9.8     | 8.4  | 6.5 | 8.3     | 6.0  | 4.2 | 8.5       | 8.5  | 8.5  | 6%      | 11% | 16% | 12%     | 23% | 33% |
| Average              |           |                                |        |              |                         |           |           |           | 71.7            | 36.4 | 24.9 | -18.6      | 0.6  | 0.6  | 10.1    | 8.1  | 6.1 | 8.4     | 6.2  | 4.7 | 26.5      | 16.7 | 12.1 | 8%      | 14% | 18% | 15%     | 26% | 31% |
| Median               |           |                                |        |              |                         |           |           |           | 75.7            | 35.9 | 18.8 | 0.3        | 0.3  | 0.2  | 10.8    | 8.9  | 6.5 | 8.3     | 6.0  | 4.2 | 31.0      | 15.9 | 9.5  | 6%      | 11% | 16% | 15%     | 23% | 33% |

Source: FactSet (for AT&S and Kinsus), Morgan Stanley Research estimates (for Unimicron, NYPGB, Zhen Ding, Ibiden and SEMCO). NC = Not covered by Morgan Stanley Research.

*Related Reports:*

[Unimicron: Very strong preliminary April profits \(27 May 2026\)](#)

[Nan Ya PCB: Strong preliminary April profits \(27 May 2026\)](#)

[Greater China Technology Hardware: ABF Substrates – AT&S announces capacity expansion in Malaysia \(15 Jun 2026\)](#)

[Nan Ya PCB: Even stronger preliminary May profits \(29 Jun 2026\)](#)

# The State of the Current Cycle

**The ABF substrate market has gone through a few cycles over the past two decades. We believe we are now still in the early innings of an upcycle.**

There was a prolonged downcycle from 2012-18 as the PC TAM shrank amid the rise of smartphones and tablets. During this tough period, companies were not able to generate profits and had to compete on price to fill their idle capacity. As a result, during those years, there was no or very limited capacity expansion, and capital expenditures by the major ABF substrate suppliers were kept at a minimum (Exhibit 3).

Even as general server demand picked up, lifting utilization rates across the group going into 2020, ABF substrate suppliers remained cautious on the longer-term demand outlook and remained disciplined, hesitating to expand capacity. Hence, capex turned up just slightly in 2019-20, increasing ~US\$600mn vs. ~US\$2.2bn on average per year in 2012-18.

This underinvestment led to sudden undersupply, as the shock from Covid-19 drove strong work-from-home demand for PCs, as well as increased data usage and storage for general computing driven by online entertainment. This ultimately led to a significant increase in ABF substrate market value from 2020-22, reflecting by the sudden surge in demand along with meaningful pricing adjustments owing to the lack of supply.

During this 2020-22 period, much capacity expansion was set in motion as ABF substrate suppliers and customers came to agree that necessary investments were needed for future growth. Some subsidies were even given to these suppliers to facilitate this capacity expansion. This is why capex jumped from ~US\$2.9bn in 2020 to US\$4.1bn in 2021 and US\$5.5bn in 2022; even in 2023-24, capex remained elevated, as capacity expansion plans continued.

Unfortunately, demand did not play out as expected. As the world came out of the pandemic, demand for PCs and online entertainment fell sharply. This normalization of demand, coupled with capacity expansion from practically every major ABF substrate supplier, resulted in the ABF substrate market once again becoming oversupplied. This led to a sharp decline in ABF substrate market value in 2023-24, as volumes normalized, driving prices down to normalize against the new supply/demand dynamic.

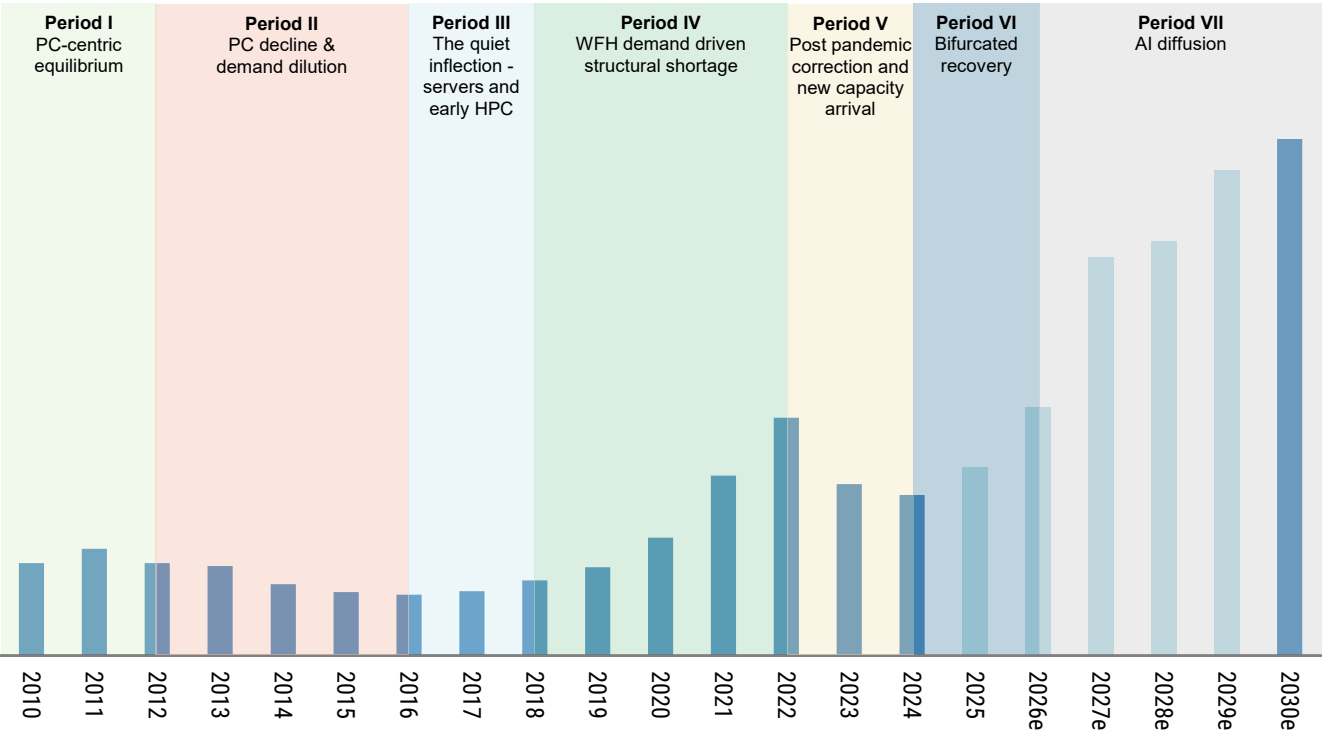
Fortunately, cloud demand remained strong for general computing, and with AI demand starting to pick up, the downcycle this time around was much shorter (two years) vs. the prior downcycle (six years).

In 1H24, the ABF substrate market troughed and we started to see an upward inflection since then, but the pace of the recovery has been moderate and bifurcated. Midrange/low-end markets such as PCs have shown moderate demand recovery, with high-end server and AI-related end-demand recovering and growing at a faster pace. This is why we turned the page on our bearish stance in 1H25, but did not turn more bullish until early 2026.

| Periods                                      | Description                                                                                                                                                                    |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Period I (2010-2012): PC-centric equilibrium | ABF substrate demand is primarily driven and dominated by CPU and GPU demand for desktops, notebooks, discrete GPUs, with a meaningfully smaller exposure towards server CPUs. |

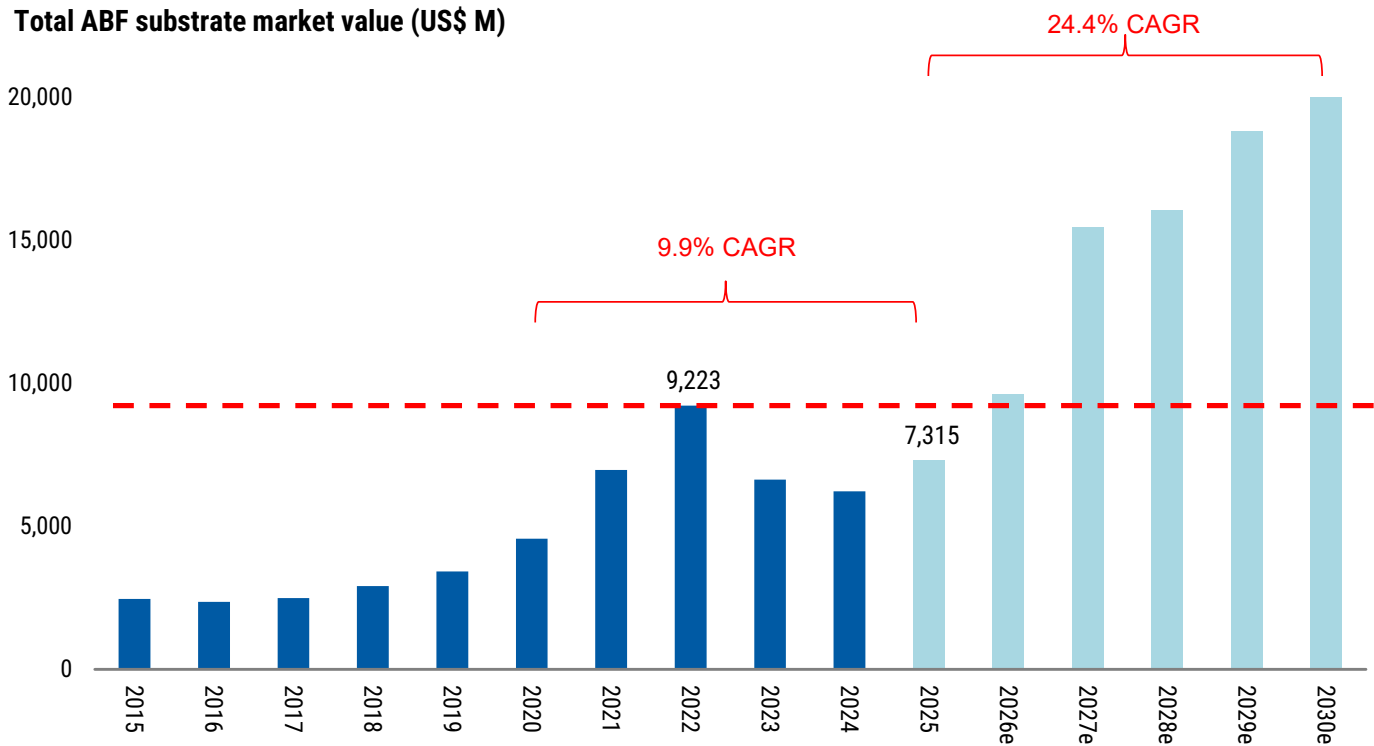
|                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Period II (2012-16): PC decline and demand dilution                     | Global PC demand entered a structural downcycle as consumer behavior changed, with smartphones and tablets absorbing more consumer time spent. This resulted in lower volumes and increased pricing competition as suppliers fought to fill idle capacity.                                                                                                                                                                                                                          |
| Period III (2016-18): The quiet inflection - servers and early HPC      | During these years, the PC market matured, while cloud computing started to take off. At the same time, server CPU substrates quietly grew larger with more layers. We also started to see early accelerators being adopted in high performance computing.                                                                                                                                                                                                                          |
| Period IV (2018-2022): WFH demand driven structural shortage            | The Covid-19 pandemic drove massive work from home demand for consumer electronics and cloud computing. This sudden surge in PC demand and increased time spent on gaming and online entertainment, bundled with a continuous increase in substrate package size and layer count, resulted in a structural shortage, which also resulted in longer lead-times and raised prices. It was during this period (2021-22) that ABF suppliers globally started to invest in new capacity. |
| Period V (2022-2024): Post-pandemic correction and new capacity arrival | As the world came out of the Covid-19 pandemic, WFH demand faded and PC and cloud computing demand were negatively affected. But at the same time, the new investments and capacity additions were just starting to slowly come online, resulting in a drop in utilization rates and pricing.                                                                                                                                                                                       |
| Period VI (2024-2026): Bifurcated recovery                              | PC demand normalized at the pre-Covid levels, with some replacement demand driven by Windows upgrade, while the recovery and growth of AI and general server demand for cloud picks up at a meaningfully higher growth trajectory.                                                                                                                                                                                                                                                  |
| Period VII (2026-30): AI diffusion                                      | Demand for AI proliferates from training to inference, driving ABF substrate market growth to re-accelerate over the next five years after a two-year downcycle following the Covid-19 pandemic. This drives increased demand for AI computing, networking, and cloud computing, while demand for PCs remains soft.                                                                                                                                                                 |

Exhibit 17: We are entering the AI diffusion era



Source: Morgan Stanley Research (e) estimates.

**Exhibit 18:** We now expect ABF substrate demand growth to accelerate, growing at 24.4% CAGR over 2025-30e (up from 22.2% previously)



Source: Prismark, Morgan Stanley Research (e) estimates.

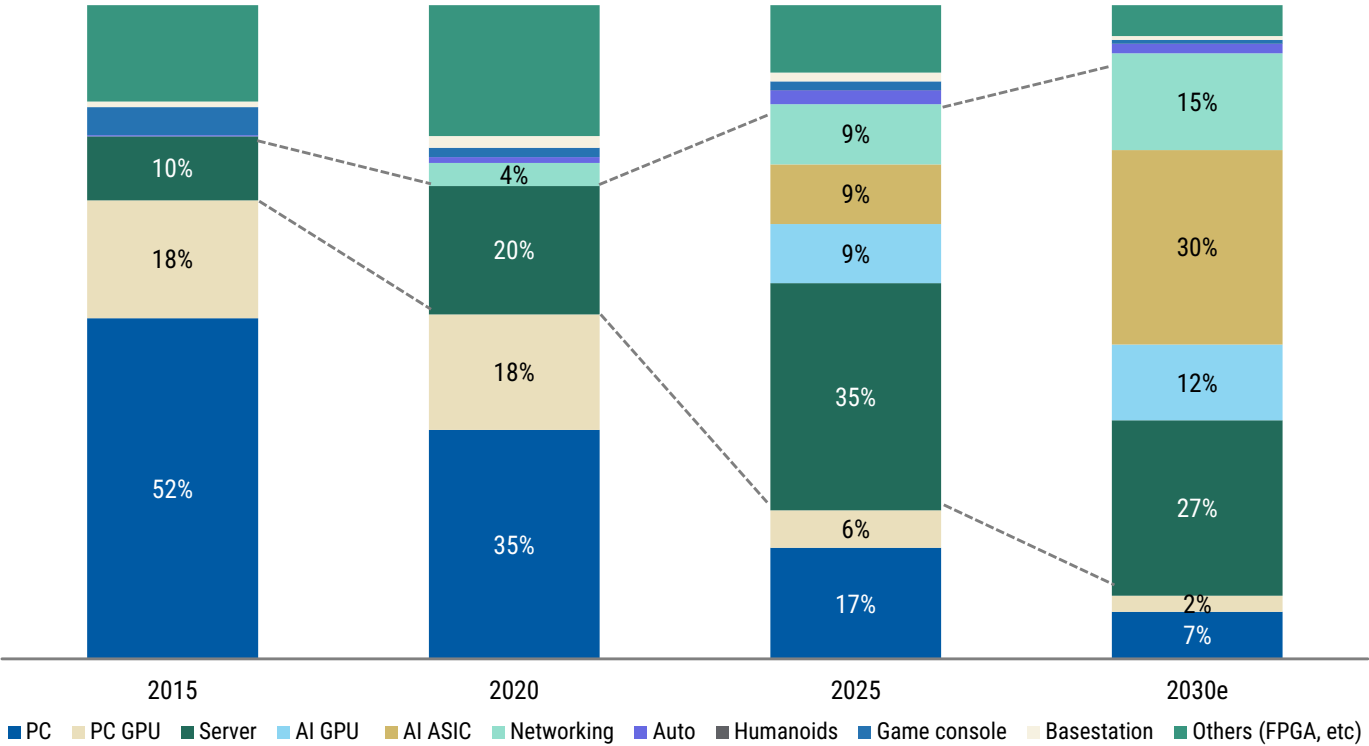
### Breaking down ABF substrate end-demand

The end-market mix for ABF substrates has shifted dramatically over the past decade. PCs were once the dominant demand driver, accounting for approximately 70% of global ABF substrate market value in 2015, supported primarily by PC CPUs and GPUs. Since then, however, the industry's demand profile has undergone a structural transformation. We estimate that PC-related applications represented only ~23% of total ABF substrate market value in 2025, and this share is expected to decline further to below 10% by 2030.

In contrast, demand from servers, AI GPUs, AI ASICs, and networking applications has expanded rapidly. We estimate these segments collectively accounted for only around 10% of the ABF substrate market in 2015, but their share increased to ~60% by 2025. Looking ahead, we expect they will represent more than 80% of total ABF substrate market value by 2030.

In our view, this structural shift in end-market demand has been the primary driver of ABF substrate industry growth over the past decade and will become even more pronounced over the next five years. As AI infrastructure deployment accelerates, we expect continued expansion in server, AI accelerator, and networking demand to drive meaningful acceleration in ABF substrate market growth.

**Exhibit 19:** PC-related ABF demand is seen declining from ~70% in 2015 to <10% by 2030, while server, AI GPU, AI ASIC, and networking-related ABF demand should rise from ~10% in 2015 to 75%+ by 2030



Source: Morgan Stanley Research (e) estimates.

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# Unimicron: Estimate Changes

We raise our 2026, 2027 and 2028 EPS estimates by 25%, 33% and 14%, respectively, primarily driven by higher assumptions for ASICs and GPUs volumes and better margin profiles.

## Exhibit 20: Estimate Changes

| Year to Dec. 31 (NT\$m)   | 2026E          |                |            | 2027E          |                |            | 2028E          |                |            |
|---------------------------|----------------|----------------|------------|----------------|----------------|------------|----------------|----------------|------------|
|                           | New            | Old            | % diff     | New            | Old            | % diff     | New            | Old            | % diff     |
| <b>P&amp;L Summary</b>    |                |                |            |                |                |            |                |                |            |
| <b>Net sales</b>          | <b>184,755</b> | <b>184,777</b> | <b>0%</b>  | <b>265,117</b> | <b>262,464</b> | <b>1%</b>  | <b>355,074</b> | <b>354,693</b> | <b>0%</b>  |
| COGS                      | (141,992)      | (147,780)      |            | (180,574)      | (193,803)      |            | (219,066)      | (233,768)      |            |
| <b>Gross profit</b>       | <b>42,763</b>  | <b>36,997</b>  | <b>16%</b> | <b>84,543</b>  | <b>68,660</b>  | <b>23%</b> | <b>136,007</b> | <b>120,924</b> | <b>12%</b> |
| Operating expenses        | (17,635)       | (17,713)       |            | (19,803)       | (19,966)       |            | (23,141)       | (21,971)       |            |
| <b>Operating income</b>   | <b>25,129</b>  | <b>19,284</b>  | <b>30%</b> | <b>64,740</b>  | <b>48,695</b>  | <b>33%</b> | <b>112,866</b> | <b>98,953</b>  | <b>14%</b> |
| Non-operating income      | 5,014          | 5,027          |            | 2,162          | 2,180          |            | 2,162          | 2,180          |            |
| <b>Pre-tax income</b>     | <b>30,143</b>  | <b>24,311</b>  | <b>24%</b> | <b>66,902</b>  | <b>50,874</b>  | <b>32%</b> | <b>115,028</b> | <b>101,133</b> | <b>14%</b> |
| <b>Net income</b>         | <b>22,855</b>  | <b>18,220</b>  | <b>25%</b> | <b>51,521</b>  | <b>38,875</b>  | <b>33%</b> | <b>89,367</b>  | <b>78,482</b>  | <b>14%</b> |
| <b>Diluted EPS (NT\$)</b> | <b>14.85</b>   | <b>11.84</b>   | <b>25%</b> | <b>33.48</b>   | <b>25.26</b>   | <b>33%</b> | <b>58.07</b>   | <b>50.99</b>   | <b>14%</b> |
| <b>Margins</b>            |                |                |            |                |                |            |                |                |            |
|                           |                |                | <i>ppt</i> |                |                | <i>ppt</i> |                |                | <i>ppt</i> |
| Gross margin              | 23.1%          | 20.0%          | 3.1        | 31.9%          | 26.2%          | 5.7        | 38.3%          | 34.1%          | 4.2        |
| Operating margin          | 13.6%          | 10.4%          | 3.2        | 24.4%          | 18.6%          | 5.9        | 31.8%          | 27.9%          | 3.9        |
| Pretax margin             | 16.3%          | 13.2%          | 3.2        | 25.2%          | 19.4%          | 5.9        | 32.4%          | 28.5%          | 3.9        |
| Net margin                | 12.4%          | 9.9%           | 2.5        | 19.4%          | 14.8%          | 4.6        | 25.2%          | 22.1%          | 3.0        |

Source: Morgan Stanley Research (E) estimates.

## Price Target and Valuation Methodology

**Our 12-month price target increases ~14% to NT\$1,465 (from NT\$1,285), driven by our increased earnings estimates for CY26-28e:** We believe the company's earnings have returned to a growth track from 2026 onward and will see stronger earnings growth in 2027, driven by the high-end ABF substrate market's reversion to undersupply and Unimicron's dominant supply share in AI ASICs and also server CPUs, which puts it in a favorable position to benefit from rising AI investments.

We believe the overall ABF substrate market bottomed in 2H24, and see growth accelerating from 2H26 onward with the ramp of multiple next-generation, large body size ABF substrates for new server GPUs, CPUs, ASICs, and networking chips. New technologies such as embedded passives and Intel's EMIB-T should also increase the manufacturing complexity of the package substrates, which adds to the under-capacity as usable output decreases.

Similar to our analysis of other tech hardware companies within our coverage, we use a residual income (RI) valuation model to value Unimicron – we think it derives the most accurate value of the firm as it takes into account cost of equity. Key assumptions are all unchanged, including:

- Cost of equity of 9.2%;
- Risk-free rate of 1% (10-year Taiwan government note yield);
- Equity risk premium of 8.7%;
- Beta of 1.0;
- Medium-term growth rate of 15%
- Terminal growth rate of 3% (similar to the rest of the companies under our tech hardware coverage).

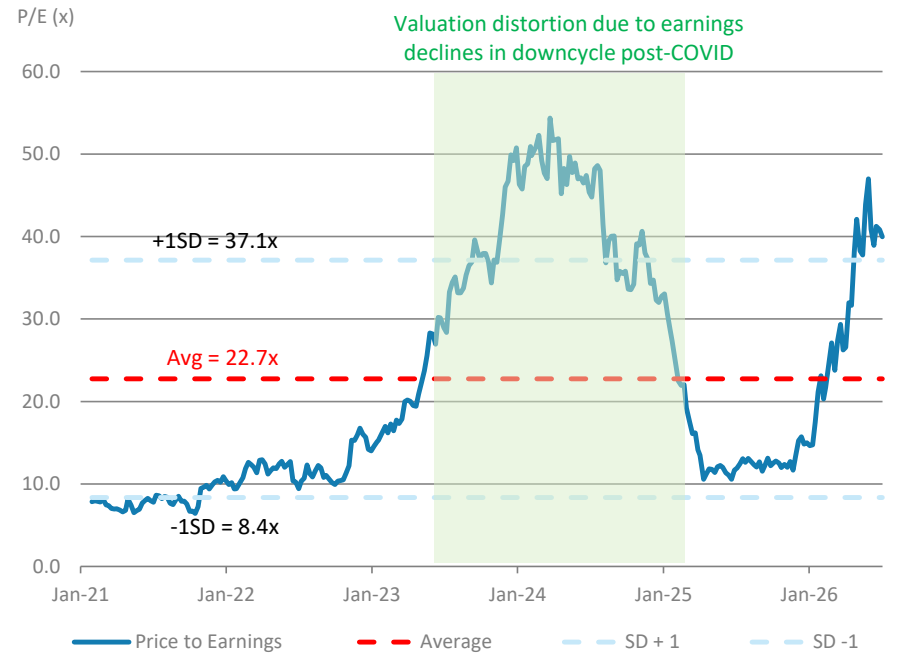
Our bull and bear case values increase by similar magnitudes, to NT\$2,500 and NT\$490, respectively.

Exhibit 21: Unimicron residual income (RI) model

|                             | 2027E     | 2028E   | 2029E   | 2030E   | 2031E   | 2032E   | 2033E   | 2034E   | 2035E   | 2036E   | 2037E   |
|-----------------------------|-----------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| Total Equity                | 167,404   | 232,927 | 271,757 | 316,412 | 367,764 | 426,820 | 494,734 | 572,835 | 662,651 | 765,940 | 851,917 |
| Net Profit                  | 51,521    | 89,367  | 102,772 | 118,188 | 135,916 | 156,304 | 179,749 | 206,712 | 237,719 | 273,376 | 281,578 |
| Return on Equity            | 35.1%     | 44.6%   | 40.7%   | 40.2%   | 39.7%   | 39.3%   | 39.0%   | 38.7%   | 38.5%   | 38.3%   | 34.8%   |
| Beta                        | 1.0       | 1.0     | 1.0     | 1.0     | 1.0     | 1.0     | 1.0     | 1.0     | 1.0     | 1.0     | 1.0     |
| Equity Risk Premium (Rm-Rf) | 8.7%      | 8.7%    | 8.7%    | 8.7%    | 8.7%    | 8.7%    | 8.7%    | 8.7%    | 8.7%    | 8.7%    | 8.7%    |
| Risk Free Rate (Rf)         | 1.0%      | 1.0%    | 1.0%    | 1.0%    | 1.0%    | 1.0%    | 1.0%    | 1.0%    | 1.0%    | 1.0%    | 1.0%    |
| Cost of Equity              | 9.2%      | 9.2%    | 9.2%    | 9.2%    | 9.2%    | 9.2%    | 9.2%    | 9.2%    | 9.2%    | 9.2%    | 9.2%    |
| Residual Income             | 32,742    | 59,377  | 73,488  | 84,275  | 96,676  | 110,935 | 127,331 | 146,185 | 167,865 | 192,796 | 196,320 |
| Spread                      | 25.9%     | 35.5%   | 31.5%   | 31.0%   | 30.6%   | 30.2%   | 29.8%   | 29.5%   | 29.3%   | 29.1%   | 25.6%   |
| Beginning Equity Capital    | 167,404   |         |         |         |         |         |         |         |         |         |         |
| PV of Forecast Period       | 726,828   |         |         |         |         |         |         |         |         |         |         |
| PV of Continuing Value      | 1,360,370 |         |         |         |         |         |         |         |         |         |         |
| Equity Value                | 2,254,602 |         |         |         |         |         |         |         |         |         |         |
| No. of Shares               | 1,539     |         |         |         |         |         |         |         |         |         |         |
| Projected Price             | 1,465.0   |         |         |         |         |         |         |         |         |         |         |

Source: Morgan Stanley Research (E) estimates.

Exhibit 22: Unimicron P/E chart



Source: TEJ, Morgan Stanley Research estimates.

## Financials



**Exhibit 23:** Quarterly Estimates

| NT\$ mn                                 | 1Q25          | 2Q25          | 3Q25          | 4Q25          | 1Q26          | 2Q26E         | 3Q26E         | 4Q26E         | 2025           | 2026E          | 2027E          | 2028E          |
|-----------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|----------------|----------------|----------------|----------------|
| <b>Sales</b>                            | <b>30,090</b> | <b>32,466</b> | <b>33,994</b> | <b>34,691</b> | <b>37,446</b> | <b>42,509</b> | <b>50,823</b> | <b>53,976</b> | <b>131,241</b> | <b>184,755</b> | <b>265,117</b> | <b>355,074</b> |
| Cost of goods sold                      | -26,063       | -28,221       | -29,446       | -29,219       | -30,720       | -33,108       | -38,335       | -39,829       | -112,949       | -141,992       | -180,574       | -219,066       |
| <b>Gross profit</b>                     | <b>4,026</b>  | <b>4,246</b>  | <b>4,549</b>  | <b>5,472</b>  | <b>6,726</b>  | <b>9,402</b>  | <b>12,488</b> | <b>14,147</b> | <b>18,292</b>  | <b>42,763</b>  | <b>84,543</b>  | <b>136,007</b> |
| Operating expenses                      | -2,759        | -2,736        | -3,016        | -3,106        | -3,969        | -4,272        | -4,654        | -4,739        | -11,618        | -17,635        | -19,803        | -23,141        |
| <b>Operating profit</b>                 | <b>1,267</b>  | <b>1,509</b>  | <b>1,533</b>  | <b>2,366</b>  | <b>2,757</b>  | <b>5,130</b>  | <b>7,834</b>  | <b>9,409</b>  | <b>6,674</b>   | <b>25,129</b>  | <b>64,740</b>  | <b>112,866</b> |
| Interest income (loss)                  | 103           | 62            | -7            | 25            | 21            | 21            | 21            | 21            | 183            | 82             | 82             | 82             |
| Investment income (loss)                | 32            | -13           | -1            | -10           | -10           | 20            | 20            | 20            | 8              | 50             | 80             | 80             |
| Gains (Losses) from investment disposal | 0             | 0             | 0             | 3             | 2             | 0             | 0             | 0             | 3              | 2              | 0              | 0              |
| Gains from FAs disposal                 | 3             | 337           | -29           | 507           | 732           | 0             | 0             | 0             | 817            | 732            | 0              | 0              |
| Exchange gain                           | 452           | -1,281        | 487           | 349           | 227           | 0             | 0             | 0             | 7              | 227            | 0              | 0              |
| Others                                  | -483          | -257          | 964           | 910           | 2,571         | 450           | 450           | 450           | 1,134          | 3,921          | 2,000          | 2,000          |
| Pre-tax profit                          | 1,373         | 357           | 2,947         | 4,150         | 6,299         | 5,620         | 8,324         | 9,899         | 8,827          | 30,143         | 66,902         | 115,028        |
| Net Income for Minority                 | 9             | 231           | 256           | 381           | 338           | 338           | 338           | 338           | 877            | 1,351          | 1,351          | 1,351          |
| Income tax                              | -450          | -97           | -496          | -235          | -918          | -1,405        | -1,831        | -1,782        | -1,277         | -5,936         | -14,030        | -24,310        |
| <b>Net profit</b>                       | <b>915</b>    | <b>30</b>     | <b>2,194</b>  | <b>3,535</b>  | <b>5,043</b>  | <b>3,877</b>  | <b>6,155</b>  | <b>7,780</b>  | <b>6,673</b>   | <b>22,855</b>  | <b>51,521</b>  | <b>89,367</b>  |
| <b>EPS (NT\$)</b>                       | <b>0.59</b>   | <b>0.02</b>   | <b>1.43</b>   | <b>2.30</b>   | <b>3.25</b>   | <b>2.50</b>   | <b>3.96</b>   | <b>5.01</b>   | <b>4.34</b>    | <b>14.85</b>   | <b>33.48</b>   | <b>58.07</b>   |
| <b>Margins (%)</b>                      |               |               |               |               |               |               |               |               |                |                |                |                |
| Gross margin                            | 13.4%         | 13.1%         | 13.4%         | 15.8%         | 18.0%         | 22.1%         | 24.6%         | 26.2%         | 13.9%          | 23.1%          | 31.9%          | 38.3%          |
| Operating margin                        | 4.2%          | 4.6%          | 4.5%          | 6.8%          | 7.4%          | 12.1%         | 15.4%         | 17.4%         | 5.1%           | 13.6%          | 24.4%          | 31.8%          |
| Pre-tax margin                          | 4.6%          | 1.1%          | 8.7%          | 12.0%         | 16.8%         | 13.2%         | 16.4%         | 18.3%         | 6.7%           | 16.3%          | 25.2%          | 32.4%          |
| Net margin                              | 3.0%          | 0.1%          | 6.5%          | 10.2%         | 13.5%         | 9.1%          | 12.1%         | 14.4%         | 5.1%           | 12.4%          | 19.4%          | 25.2%          |
| <b>QoQ Growth</b>                       |               |               |               |               |               |               |               |               |                |                |                |                |
| Sales                                   | 2%            | 8%            | 5%            | 2%            | 8%            | 14%           | 20%           | 6%            |                |                |                |                |
| Gross profit                            | 18%           | 5%            | 7%            | 20%           | 23%           | 40%           | 33%           | 13%           |                |                |                |                |
| Operating profit                        | 86%           | 19%           | 2%            | 54%           | 17%           | 86%           | 53%           | 20%           |                |                |                |                |
| Pre-tax profit                          | 331%          | -74%          | 726%          | 41%           | 52%           | -11%          | 48%           | 19%           |                |                |                |                |
| Net profit                              | 1531%         | -97%          | 7310%         | 61%           | 41%           | -23%          | 59%           | 26%           |                |                |                |                |
| <b>YoY Growth</b>                       |               |               |               |               |               |               |               |               |                |                |                |                |
| Sales                                   | 14%           | 16%           | 7%            | 18%           | 24%           | 31%           | 50%           | 56%           | 14%            | 41%            | 43%            | 34%            |
| Gross profit                            | -6%           | 15%           | -8%           | 61%           | 67%           | 121%          | 175%          | 159%          | 12%            | 134%           | 98%            | 61%            |
| Operating profit                        | -20%          | 68%           | -22%          | 248%          | 118%          | 240%          | 411%          | 298%          | 30%            | 276%           | 158%           | 74%            |
| Pre-tax profit                          | -59%          | -83%          | 95%           | 1201%         | 359%          | 1475%         | 182%          | 139%          | 21%            | 241%           | 122%           | 72%            |
| Net profit                              | -62%          | -98%          | 120%          | 6204%         | 451%          | 12996%        | 180%          | 120%          | 31%            | 242%           | 125%           | 73%            |

Source: Company data, Morgan Stanley Research (E) estimates.

**Exhibit 24: Consolidated Financial Summary****Income Statement**

| NT\$ mn; FY End Dec              | 2025           | 2026E          | 2027E          | 2028E          |
|----------------------------------|----------------|----------------|----------------|----------------|
| <b>Net sales</b>                 | <b>131,241</b> | <b>184,755</b> | <b>265,117</b> | <b>355,074</b> |
| Cost of goods sold               | -112,949       | -141,992       | -180,574       | -219,066       |
| <b>Gross profit</b>              | <b>18,292</b>  | <b>42,763</b>  | <b>84,543</b>  | <b>136,007</b> |
| Operating expenses               | -11,618        | -17,635        | -19,803        | -23,141        |
| Sales & Marketing                | -1,607         | -2,485         | -3,216         | -3,919         |
| G&A                              | -5,012         | -7,650         | -7,877         | -9,078         |
| R&D                              | -4,999         | -7,499         | -8,710         | -10,144        |
| <b>Operating profit</b>          | <b>6,674</b>   | <b>25,129</b>  | <b>64,740</b>  | <b>112,866</b> |
| Interest income                  | 183            | 82             | 82             | 82             |
| Investment income                | 8              | 50             | 80             | 80             |
| Gains from investment disposal   | 3              | 2              | 0              | 0              |
| Gains from fixed assets disposal | 817            | 732            | 0              | 0              |
| Exchange gain                    | 7              | 227            | 0              | 0              |
| Other non-operating income       | 1,134          | 3,921          | 2,000          | 2,000          |
| <b>Net non-operating profit</b>  | <b>2,153</b>   | <b>5,014</b>   | <b>2,162</b>   | <b>2,162</b>   |
| Pre-tax profit                   | 8,827          | 30,143         | 66,902         | 115,028        |
| Net Income for Minority          | -877           | -1,351         | -1,351         | -1,351         |
| Income tax                       | -1,277         | -5,936         | -14,030        | -24,310        |
| <b>Net profit</b>                | <b>6,673</b>   | <b>22,855</b>  | <b>51,521</b>  | <b>89,367</b>  |
| <b>Reported EPS (NT\$)</b>       | <b>4.34</b>    | <b>14.85</b>   | <b>33.48</b>   | <b>58.07</b>   |

**Balance Sheet**

| NT\$ mn; FY End Dec                 | 2025           | 2026E          | 2027E          | 2028E          |
|-------------------------------------|----------------|----------------|----------------|----------------|
| Cash                                | 54,872         | 84,130         | 151,335        | 243,258        |
| Mkt Securities                      | 38             | 38             | 38             | 38             |
| AR/NR                               | 27,570         | 38,812         | 55,693         | 74,591         |
| Inventory                           | 17,802         | 22,379         | 28,460         | 34,527         |
| Others                              | 5,016          | 7,061          | 10,132         | 13,570         |
| <b>Current Assets</b>               | <b>105,298</b> | <b>152,420</b> | <b>245,660</b> | <b>365,984</b> |
| Long-term Investments               | 836            | 836            | 836            | 836            |
| Fixed Assets                        | 125,866        | 129,535        | 121,811        | 114,387        |
| Other Assets                        | 23,792         | 23,792         | 23,792         | 23,792         |
| <b>Total Assets</b>                 | <b>255,792</b> | <b>306,584</b> | <b>392,099</b> | <b>505,000</b> |
| S-t Borrowings                      | 10,845         | 10,845         | 10,845         | 10,845         |
| AP/NP                               | 17,494         | 21,993         | 27,969         | 33,931         |
| Other S-T Liabilities               | 47,115         | 59,231         | 75,325         | 91,382         |
| Other LT Liabilities                | 38,457         | 54,138         | 77,686         | 104,045        |
| L-t Debt                            | 34,873         | 33,872         | 32,871         | 31,870         |
| <b>Total Liabilities</b>            | <b>148,785</b> | <b>180,078</b> | <b>224,695</b> | <b>272,073</b> |
| Common Shares                       | 15,296         | 15,296         | 15,296         | 15,296         |
| Other Shareholders' Equity          | 91,711         | 111,209        | 152,108        | 217,631        |
| <b>Total Equity</b>                 | <b>107,007</b> | <b>126,505</b> | <b>167,404</b> | <b>232,927</b> |
| <b>Total Liab./Shrhldr's Equity</b> | <b>255,792</b> | <b>306,584</b> | <b>392,099</b> | <b>505,000</b> |

**Cash Flow Statement**

| NT\$ mn; FY End Dec               | 2025           | 2026E          | 2027E          | 2028E          |
|-----------------------------------|----------------|----------------|----------------|----------------|
| <b>Cashflow from Operations</b>   | <b>14,967</b>  | <b>41,439</b>  | <b>67,555</b>  | <b>101,785</b> |
| Net Profits                       | 6,673          | 22,855         | 51,521         | 89,367         |
| Depreciation                      | 18,773         | 19,887         | 20,078         | 18,881         |
| Investment losses (income)        | 0              | -50            | -80            | -80            |
| Investment Disposal Loss          | -3             | -2             | 0              | 0              |
| Change in Working Capital         | -2,943         | 794            | -893           | -2,945         |
| Other adjustments                 | -7,533         | -2,045         | -3,071         | -3,438         |
| <b>Cashflow from Investing</b>    | <b>-23,018</b> | <b>-23,556</b> | <b>-12,354</b> | <b>-11,456</b> |
| (Purchases) Sale of fixed asset   | -24,307        | -23,556        | -12,354        | -11,456        |
| (Purchases) Sale of LT investment | -77            | 0              | 0              | 0              |
| Other adjustments                 | 1,367          | 0              | 0              | 0              |
| <b>Cashflow from financing</b>    | <b>19,073</b>  | <b>11,602</b>  | <b>12,005</b>  | <b>1,594</b>   |
| Increase in L-T debt              | 9,950          | -1,001         | -1,001         | -1,001         |
| Increase in S-T debt              | 4,331          | 0              | 0              | 0              |
| Issuance of stock                 | 0              | 0              | 0              | 0              |
| Other adjustments                 | 7,087          | 15,681         | 23,548         | 26,360         |
| Exchange Rate Adjustment          | 96             | -227           | 0              | 0              |
| Net change in cash                | 11,118         | 29,258         | 67,206         | 91,922         |

**Financial Ratios**

|                               | 2025   | 2026E  | 2027E  | 2028E  |
|-------------------------------|--------|--------|--------|--------|
| <b>Margins</b>                |        |        |        |        |
| Gross margin                  | 13.9%  | 23.1%  | 31.9%  | 38.3%  |
| Operating expenses/sales      | -8.9%  | -9.5%  | -7.5%  | -6.5%  |
| Operating margin              | 5.1%   | 13.6%  | 24.4%  | 31.8%  |
| Tax rate                      | 14.5%  | 19.7%  | 21.0%  | 21.1%  |
| Pre-tax margin                | 6.7%   | 16.3%  | 25.2%  | 32.4%  |
| Net Margin                    | 5.1%   | 12.4%  | 19.4%  | 25.2%  |
| <b>YoY Growth</b>             |        |        |        |        |
| Turnover                      | 13.8%  | 40.8%  | 43.5%  | 33.9%  |
| Gross Profit                  | 12.1%  | 133.8% | 97.7%  | 60.9%  |
| Operating Profits             | 30.4%  | 276.5% | 157.6% | 74.3%  |
| Pretax Profits                | 20.6%  | 241.5% | 122.0% | 71.9%  |
| Net Profits                   | 31.3%  | 242.5% | 125.4% | 73.5%  |
| Adjusted Cash Dividend (NT\$) | 2.00   | 6.85   | 15.44  | 26.78  |
| Net Debt/Equity               | -8.6%  | -31.2% | -64.3% | -86.1% |
| Liabilities/Equity            | 139.0% | 142.3% | 134.2% | 116.8% |
| Liabilities/Assets            | 58.2%  | 58.7%  | 57.3%  | 53.9%  |
| ROAE                          | 6.5%   | 19.6%  | 35.1%  | 44.6%  |
| ROAA                          | 2.7%   | 8.1%   | 14.7%  | 19.9%  |
| A/R Turnover (days)           | 71.2   | 65.6   | 65.1   | 67.0   |
| A/P Turnover (days)           | 52.7   | 50.8   | 50.5   | 51.6   |
| Inventory Turnover (days)     | 52.8   | 51.6   | 51.4   | 52.5   |
| Cash conversion (days)        | 71.3   | 66.5   | 65.9   | 67.9   |

Source: Company data, Morgan Stanley Research (E) estimates.

# NYPCB: Estimate Changes

We raise our 2026, 2027 and 2028 EPS estimates by 36%, 39% and 23%, respectively, primarily driven by higher assumptions for AI semiconductors volumes and higher margin profiles.

## Exhibit 25: Estimate Changes

| NT\$ mn                 | 2026E         |               |            | 2027E         |               |            | 2028E          |                |            |
|-------------------------|---------------|---------------|------------|---------------|---------------|------------|----------------|----------------|------------|
|                         | New           | Old           | Variance   | New           | Old           | Variance   | New            | Old            | Variance   |
| <b>Net sales</b>        | <b>56,808</b> | <b>55,058</b> | <b>3%</b>  | <b>78,966</b> | <b>74,811</b> | <b>6%</b>  | <b>111,555</b> | <b>109,356</b> | <b>2%</b>  |
| COGS                    | -43,726       | -45,166       | -3%        | -52,309       | -55,294       | -5%        | -61,587        | -68,495        | -10%       |
| <b>Gross profit</b>     | <b>13,081</b> | <b>9,891</b>  | <b>32%</b> | <b>26,657</b> | <b>19,517</b> | <b>37%</b> | <b>49,968</b>  | <b>40,861</b>  | <b>22%</b> |
| Operating expenses      | -1,805        | -1,788        | 1%         | -2,045        | -2,006        | 2%         | -2,397         | -2,377         | 1%         |
| <b>Operating income</b> | <b>11,276</b> | <b>8,103</b>  | <b>39%</b> | <b>24,612</b> | <b>17,511</b> | <b>41%</b> | <b>47,572</b>  | <b>38,484</b>  | <b>24%</b> |
| Non-operating income    | 811           | 811           | 0%         | 762           | 762           | 0%         | 762            | 762            | 0%         |
| <b>Pre-tax income</b>   | <b>12,087</b> | <b>8,914</b>  | <b>36%</b> | <b>25,374</b> | <b>18,273</b> | <b>39%</b> | <b>48,333</b>  | <b>39,245</b>  | <b>23%</b> |
| Income tax              | -1,979        | -1,472        | 34%        | -4,037        | -2,905        | 39%        | -7,712         | -6,256         | 23%        |
| <b>Net income</b>       | <b>10,108</b> | <b>7,442</b>  | <b>36%</b> | <b>21,337</b> | <b>15,367</b> | <b>39%</b> | <b>40,621</b>  | <b>32,989</b>  | <b>23%</b> |
| <b>EPS (NT\$)</b>       | <b>15.64</b>  | <b>11.52</b>  | <b>36%</b> | <b>33.02</b>  | <b>23.78</b>  | <b>39%</b> | <b>62.87</b>   | <b>51.05</b>   | <b>23%</b> |
| <b>Margins (%)</b>      |               |               | <b>ppt</b> |               |               | <b>ppt</b> |                |                | <b>ppt</b> |
| Gross margin            | 23.0%         | 18.0%         | 5.1        | 33.8%         | 26.1%         | 7.7        | 44.8%          | 37.4%          | 7.4        |
| Operating margin        | 19.9%         | 14.7%         | 5.1        | 31.2%         | 23.4%         | 7.8        | 42.6%          | 35.2%          | 7.5        |
| Pre-tax margin          | 21.3%         | 16.2%         | 5.1        | 32.1%         | 24.4%         | 7.7        | 43.3%          | 35.9%          | 7.4        |
| Net margin              | 17.8%         | 13.5%         | 4.3        | 27.0%         | 20.5%         | 6.5        | 36.4%          | 30.2%          | 6.2        |

Source: Morgan Stanley Research (E) estimates.

## Price Target and Valuation Methodology

### Our price target increases ~22% to NT\$1,550, on the back of our raised earnings

**estimates for CY26-28e:** We believe the company's earnings have returned to a growth track from 2026 onward and will see stronger growth in 2027, driven by the high-end ABF substrate market's reversion to undersupply, NYPCB's dominant supply share for networking chips, and its proactive pricing strategy, which puts it in a favorable position to benefit from rising AI investments.

We believe the overall ABF substrate market bottomed in 2H24, and growth should accelerate from 2H26 onward with the ramp of multiple next-generation, large body size ABF substrates for new server GPUs, CPUs, ASICs and networking chips. New technologies, such as embedded passives and Intel's EMIB-T, should also increase manufacturing complexity of the package substrates, adding to the under-capacity as usable output decreases.

Similar to our analysis of other tech hardware companies within our coverage, we use a residual income (RI) valuation model to value NYPCB – we think it derives the most accurate value of the firm given that it takes into account cost of equity. Key assumptions (all unchanged):

- Cost of equity of 9.3%;
- Risk-free rate of 1% (10-year Taiwan government note yield);
- Equity risk premium of 8.7%;
- Beta of 1.0;
- Medium-term growth rate of 17% - fair given our analysis of the substrate market over the next couple of years;
- Terminal growth rate of 3% (similar to the rest of the companies under our tech hardware coverage).

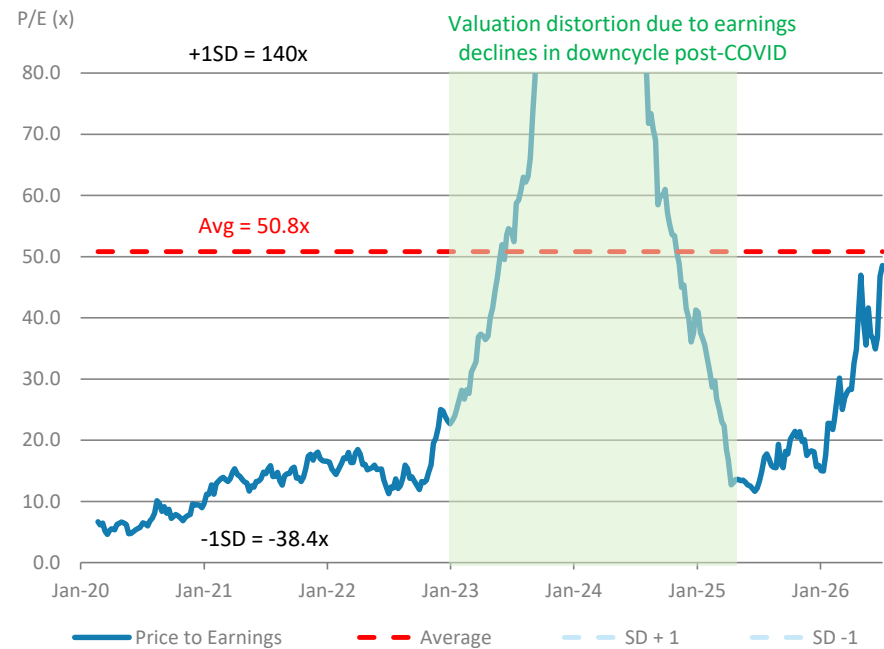
Our bull and bear case values increase by similar magnitudes, to NT\$2,770 and NT\$790, respectively.

Exhibit 26: NYPCB residual income (RI) model

|                             | 2027E     | 2028E  | 2029E   | 2030E   | 2031E   | 2032E   | 2033E   | 2034E   | 2035E   | 2036E   | 2037E   |
|-----------------------------|-----------|--------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| Total Equity                | 69,401    | 95,832 | 124,068 | 157,103 | 195,755 | 240,978 | 293,888 | 355,794 | 428,223 | 512,965 | 592,144 |
| Net Profit                  | 21,337    | 40,621 | 47,527  | 55,606  | 65,059  | 76,119  | 89,060  | 104,200 | 121,914 | 142,639 | 146,918 |
| Return on Equity            | 34.4%     | 49.2%  | 43.2%   | 39.6%   | 36.9%   | 34.9%   | 33.3%   | 32.1%   | 31.1%   | 30.3%   | 26.6%   |
| Beta                        | 1.0       | 1.0    | 1.0     | 1.0     | 1.0     | 1.0     | 1.0     | 1.0     | 1.0     | 1.0     | 1.0     |
| Equity Risk Premium (Rm-Rf) | 8.7%      | 8.7%   | 8.7%    | 8.7%    | 8.7%    | 8.7%    | 8.7%    | 8.7%    | 8.7%    | 8.7%    | 8.7%    |
| Risk Free Rate (Rf)         | 1.0%      | 1.0%   | 1.0%    | 1.0%    | 1.0%    | 1.0%    | 1.0%    | 1.0%    | 1.0%    | 1.0%    | 1.0%    |
| Cost of Equity              | 9.3%      | 9.3%   | 9.3%    | 9.3%    | 9.3%    | 9.3%    | 9.3%    | 9.3%    | 9.3%    | 9.3%    | 9.3%    |
| Residual Income             | 13,752    | 27,694 | 32,546  | 37,579  | 43,379  | 50,103  | 57,925  | 67,045  | 77,690  | 90,126  | 88,871  |
| Spread                      | 25.1%     | 39.9%  | 34.0%   | 30.3%   | 27.6%   | 25.6%   | 24.0%   | 22.8%   | 21.8%   | 21.0%   | 17.3%   |
| Beginning Equity Capital    | 69,401    |        |         |         |         |         |         |         |         |         |         |
| PV of Forecast Period       | 329,611   |        |         |         |         |         |         |         |         |         |         |
| PV of Continuing Value      | 602,522   |        |         |         |         |         |         |         |         |         |         |
| Equity Value                | 1,001,534 |        |         |         |         |         |         |         |         |         |         |
| No. of Shares               | 646       |        |         |         |         |         |         |         |         |         |         |
| Projected Price             | 1,550.0   |        |         |         |         |         |         |         |         |         |         |

Source: Morgan Stanley Research (E) estimates.

Exhibit 27: NYPCB P/E chart



Source: TEJ, Morgan Stanley Research estimates.

Financials

## Exhibit 28: Quarterly Estimates

|                             | 1Q25         | 2Q25          | 3Q25          | 4Q25          | 1Q26          | 2Q26E         | 3Q26E         | 4Q26E         | 2025          | 2026E         | 2027E         | 2028E          |
|-----------------------------|--------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|----------------|
| <b>Net sales</b>            | <b>8,458</b> | <b>9,583</b>  | <b>10,967</b> | <b>11,165</b> | <b>11,177</b> | <b>13,575</b> | <b>15,552</b> | <b>16,504</b> | <b>40,173</b> | <b>56,808</b> | <b>78,966</b> | <b>111,555</b> |
| COGS                        | (8,028)      | (8,814)       | (9,918)       | (9,766)       | (9,406)       | (10,666)      | (11,599)      | (12,056)      | (36,526)      | (43,726)      | (52,309)      | (61,587)       |
| <b>Gross profit</b>         | <b>430</b>   | <b>769</b>    | <b>1,050</b>  | <b>1,398</b>  | <b>1,771</b>  | <b>2,909</b>  | <b>3,953</b>  | <b>4,448</b>  | <b>3,647</b>  | <b>13,081</b> | <b>26,657</b> | <b>49,968</b>  |
| Operating expenses          | (397)        | (403)         | (429)         | (435)         | (419)         | (445)         | (467)         | (474)         | (1,665)       | (1,805)       | (2,045)       | (2,397)        |
| SG&A                        | (397)        | (403)         | (429)         | (435)         | (419)         | (445)         | (467)         | (474)         | (1,665)       | (1,805)       | (2,045)       | (2,397)        |
| R&D                         | 0            | 0             | 0             | 0             | 0             | 0             | 0             | 0             | 0             | 0             | 0             | 0              |
| <b>Operating income</b>     | <b>33</b>    | <b>366</b>    | <b>620</b>    | <b>963</b>    | <b>1,352</b>  | <b>2,464</b>  | <b>3,486</b>  | <b>3,974</b>  | <b>1,982</b>  | <b>11,276</b> | <b>24,612</b> | <b>47,572</b>  |
| <b>Non-operating income</b> | <b>223</b>   | <b>(593)</b>  | <b>287</b>    | <b>446</b>    | <b>246</b>    | <b>176</b>    | <b>193</b>    | <b>196</b>    | <b>363</b>    | <b>811</b>    | <b>762</b>    | <b>762</b>     |
| Interest income             | 45           | 47            | 19            | 34            | 34            | 34            | 34            | 34            | 144           | 135           | 135           | 135            |
| Investment income           | 5            | (8)           | 9             | 12            | 13            | (8)           | 9             | 12            | 18            | 26            | 26            | 26             |
| Disposal of investment      | 0            | 0             | 0             | 0             | 0             | 0             | 0             | 0             | 0             | 0             | 0             | 0              |
| Exchange gain               | 124          | (699)         | 167           | 249           | 112           | 0             | 0             | 0             | (158)         | 112           | 0             | 0              |
| Other                       | 49           | 67            | 91            | 151           | 87            | 150           | 150           | 150           | 358           | 537           | 600           | 600            |
| <b>Pre-tax income</b>       | <b>256</b>   | <b>(227)</b>  | <b>907</b>    | <b>1,409</b>  | <b>1,598</b>  | <b>2,640</b>  | <b>3,679</b>  | <b>4,170</b>  | <b>2,346</b>  | <b>12,087</b> | <b>25,374</b> | <b>48,333</b>  |
| Income tax                  | (49)         | 39            | (182)         | (207)         | (290)         | (475)         | (589)         | (625)         | (399)         | (1,979)       | (4,037)       | (7,712)        |
| Minority interest           | 0            | 0             | 0             | 0             | 0             | 0             | 0             | 0             | 0             | 0             | 0             | 0              |
| <b>Net income</b>           | <b>207</b>   | <b>(187)</b>  | <b>725</b>    | <b>1,202</b>  | <b>1,309</b>  | <b>2,165</b>  | <b>3,091</b>  | <b>3,544</b>  | <b>1,947</b>  | <b>10,108</b> | <b>21,337</b> | <b>40,621</b>  |
| Adj.wtd.avg.shrs (mn)       | 646          | 646           | 646           | 646           | 646           | 646           | 646           | 646           | 646           | 646           | 646           | 646            |
| <b>EPS</b>                  | <b>0.32</b>  | <b>(0.29)</b> | <b>1.12</b>   | <b>1.86</b>   | <b>2.03</b>   | <b>3.35</b>   | <b>4.78</b>   | <b>5.48</b>   | <b>3.01</b>   | <b>15.64</b>  | <b>33.02</b>  | <b>62.87</b>   |
| Fully Diluted shares (mn)   | 646          | 646           | 646           | 646           | 646           | 646           | 646           | 646           | 646           | 646           | 646           | 646            |
| <b>Diluted EPS (NT\$)</b>   | <b>0.32</b>  | <b>(0.29)</b> | <b>1.12</b>   | <b>1.86</b>   | <b>2.03</b>   | <b>3.35</b>   | <b>4.78</b>   | <b>5.48</b>   | <b>3.01</b>   | <b>15.64</b>  | <b>33.02</b>  | <b>62.87</b>   |
| <b>Margins (%)</b>          |              |               |               |               |               |               |               |               |               |               |               |                |
| Gross Margin                | 5.1          | 8.0           | 9.6           | 12.5          | 15.8          | 21.4          | 25.4          | 27.0          | 9.1           | 23.0          | 33.8          | 44.8           |
| Operating Margin            | 0.4          | 3.8           | 5.7           | 8.6           | 12.1          | 18.2          | 22.4          | 24.1          | 4.9           | 19.9          | 31.2          | 42.6           |
| Pretax Margin               | 3.0          | (2.4)         | 8.3           | 12.6          | 14.3          | 19.4          | 23.7          | 25.3          | 5.8           | 21.3          | 32.1          | 43.3           |
| Net Margin                  | 2.5          | (2.0)         | 6.6           | 10.8          | 11.7          | 15.9          | 19.9          | 21.5          | 4.8           | 17.8          | 27.0          | 36.4           |
| <b>QoQ Growth (%)</b>       |              |               |               |               |               |               |               |               |               |               |               |                |
| Sales                       | 7            | 13            | 14            | 2             | 0             | 21            | 15            | 6             |               |               |               |                |
| Gross Profit                | 2,441        | 79            | 37            | 33            | 27            | 64            | 36            | 13            |               |               |               |                |
| Operating Profit            | NM           | 10            | 69            | 55            | 40            | 82            | 41            | 14            |               |               |               |                |
| Pretax Profit               | 63           | -188          | NM            | 55            | 13            | 65            | 39            | 13            |               |               |               |                |
| Net Profit                  | 16           | -190          | NM            | 66            | 9             | 65            | 43            | 15            |               |               |               |                |
| <b>YoY Growth (%)</b>       |              |               |               |               |               |               |               |               |               |               |               |                |
| Sales                       | 19           | 18            | 19            | 42            | 32            | 42            | 42            | 48            | 24            | 41            | 39            | 41             |
| Gross Profit                | NM           | 319           | 95            | 8,157         | 312           | 278           | 277           | 218           | 920           | 259           | 104           | 87             |
| Operating Profit            | NM           | NM            | 396           | NM            | 3,986         | 573           | 462           | 313           | NM            | 469           | 118           | 93             |
| Pretax Profit               | NM           | -226          | 1,247         | 794           | 524           | NM            | 306           | 196           | 1,338         | 415           | 110           | 90             |
| Net Profit                  | NM           | -258          | 1,131         | 573           | 531           | NM            | 326           | 195           | 856           | 419           | 111           | 90             |

Source: Company data, Morgan Stanley Research (E) estimates.

**Exhibit 29: Consolidated Financial Summary****Consolidated Income Statement**

| NT\$mnn (Year End Dec 31)   | 2025          | 2026E         | 2027E         | 2028E          |
|-----------------------------|---------------|---------------|---------------|----------------|
| <b>Net sales</b>            | <b>40,173</b> | <b>56,808</b> | <b>78,966</b> | <b>111,555</b> |
| COGS                        | (36,526)      | (43,726)      | (52,309)      | (61,587)       |
| <b>Gross profit</b>         | <b>3,647</b>  | <b>13,081</b> | <b>26,657</b> | <b>49,968</b>  |
| Operating expenses          | (1,665)       | (1,805)       | (2,045)       | (2,397)        |
| <b>Operating income</b>     | <b>1,982</b>  | <b>11,276</b> | <b>24,612</b> | <b>47,572</b>  |
| <b>Non-operating income</b> | <b>363</b>    | <b>811</b>    | <b>762</b>    | <b>762</b>     |
| Interest income             | 144           | 135           | 135           | 135            |
| Investment income           | 18            | 26            | 26            | 26             |
| Disposal of investment      | -             | -             | -             | -              |
| Exchange gain               | (158)         | 112           | -             | -              |
| Other                       | 358           | 537           | 600           | 600            |
| <b>Pre-tax income</b>       | <b>2,346</b>  | <b>12,087</b> | <b>25,374</b> | <b>48,333</b>  |
| Income tax                  | (399)         | (1,979)       | (4,037)       | (7,712)        |
| Minority interests          | -             | -             | -             | -              |
| <b>Net income</b>           | <b>1,947</b>  | <b>10,108</b> | <b>21,337</b> | <b>40,621</b>  |
| Adj. wtd. Avg. shrs (m)     | 646           | 646           | 646           | 646            |
| <b>Reported EPS (NT\$)</b>  | <b>3.01</b>   | <b>15.64</b>  | <b>33.02</b>  | <b>62.87</b>   |
| Diluted shrs (m)            | 646           | 646           | 646           | 646            |
| <b>Diluted EPS (NT\$)</b>   | <b>3.01</b>   | <b>15.64</b>  | <b>33.02</b>  | <b>62.87</b>   |

**Consolidated Balance Sheet**

| NT\$mnn (Year End Dec 31)         | 2025          | 2026E         | 2027E         | 2028E          |
|-----------------------------------|---------------|---------------|---------------|----------------|
| <b>Cash</b>                       | <b>8,590</b>  | <b>19,996</b> | <b>36,648</b> | <b>64,276</b>  |
| Mkt securities                    | 117           | 117           | 117           | 117            |
| AR/NR                             | 9,552         | 13,508        | 18,776        | 26,526         |
| Inventory                         | 5,068         | 6,067         | 7,257         | 8,545          |
| Others                            | 296           | 419           | 582           | 823            |
| <b>Current Assets</b>             | <b>23,623</b> | <b>40,106</b> | <b>63,381</b> | <b>100,286</b> |
| Long-term investments             | 500           | 500           | 500           | 500            |
| Fixed assets                      | 35,373        | 32,254        | 29,624        | 27,381         |
| Other assets                      | 1,711         | 1,711         | 1,711         | 1,711          |
| <b>Total Assets</b>               | <b>61,207</b> | <b>74,572</b> | <b>95,217</b> | <b>129,878</b> |
| S/T borrowings                    | -             | -             | -             | -              |
| AP/NP                             | 1,711         | 2,048         | 2,450         | 2,885          |
| Other ST liabilities              | 5,478         | 6,558         | 7,845         | 9,236          |
| <b>Total Current Liabilities</b>  | <b>7,189</b>  | <b>8,606</b>  | <b>10,295</b> | <b>12,121</b>  |
| L/T debt                          | -             | -             | -             | -              |
| Other LT liabilities              | 7,896         | 11,165        | 15,520        | 21,925         |
| <b>Total Liabilities</b>          | <b>15,085</b> | <b>19,771</b> | <b>25,815</b> | <b>34,047</b>  |
| Common shares                     | 6,462         | 6,462         | 6,462         | 6,462          |
| Retained Earnings                 | 13,031        | 21,709        | 36,309        | 62,740         |
| Other SH' Equity                  | 26,630        | 26,630        | 26,630        | 26,630         |
| <b>Total Shareholders' Equity</b> | <b>46,123</b> | <b>54,801</b> | <b>69,401</b> | <b>95,832</b>  |
| <b>Total Liab./SH's Equity</b>    | <b>61,207</b> | <b>74,572</b> | <b>95,217</b> | <b>129,878</b> |

**Consolidated Cash Flow Statement**

| NT\$mnn (Year End Dec 31)  | 2025           | 2026E          | 2027E          | 2028E          |
|----------------------------|----------------|----------------|----------------|----------------|
| <b>Operating Cashflow</b>  | <b>3,027</b>   | <b>12,436</b>  | <b>21,860</b>  | <b>38,180</b>  |
| Net Profits                | 1,947          | 10,108         | 21,337         | 40,621         |
| Depreciation & Amort.      | 6,894          | 6,013          | 5,483          | 5,036          |
| Investment losses/(income) | (18)           | (26)           | (26)           | (26)           |
| Working capital change     | 12,615         | (3,537)        | (4,770)        | (7,210)        |
| Other adjustments          | (18,410)       | (123)          | (163)          | (240)          |
| <b>Investing Cashflow</b>  | <b>(1,723)</b> | <b>(2,894)</b> | <b>(2,853)</b> | <b>(2,793)</b> |
| Capex                      | (2,422)        | (3,000)        | (3,000)        | (3,000)        |
| Change of L/T investment   | 0              | 0              | 0              | 0              |
| Change of S/T investment   | 0              | 0              | 0              | 0              |
| Other adjustments          | 625            | 0              | 0              | 0              |
| <b>Financing Cashflow</b>  | <b>(887)</b>   | <b>1,977</b>   | <b>(2,355)</b> | <b>(7,759)</b> |
| Increase in L/T debt       | 0              | 0              | 0              | 0              |
| Increase in S/T debt       | 0              | 0              | 0              | 0              |
| Issuance of stock          | 0              | (0)            | 0              | 0              |
| Cash dividends             | (646)          | (1,292)        | (6,710)        | (14,164)       |
| Other adjustments          | (241)          | 3,269          | 4,355          | 6,405          |
| FX adjustment              | (309)          | (112)          | 0              | 0              |
| <b>Net change in cash</b>  | <b>109</b>     | <b>11,406</b>  | <b>16,652</b>  | <b>27,628</b>  |

**Consolidated Financial Ratios**

|                           | 2025    | 2026E | 2027E | 2028E |
|---------------------------|---------|-------|-------|-------|
| <b>Margins (%)</b>        |         |       |       |       |
| Gross margin              | 9.1     | 23.0  | 33.8  | 44.8  |
| Operating margin          | 4.9     | 19.9  | 31.2  | 42.6  |
| Pretax margin             | 5.8     | 21.3  | 32.1  | 43.3  |
| Net margin                | 4.8     | 17.8  | 27.0  | 36.4  |
| <b>YoY growth (%)</b>     |         |       |       |       |
| Sales                     | 24.4    | 41.4  | 39.0  | 41.3  |
| Operating profits         | NM      | 468.8 | 118.3 | 93.3  |
| Pretax profits            | 1,338.4 | 415.3 | 109.9 | 90.5  |
| Net profits               | 855.6   | 419.2 | 111.1 | 90.4  |
| <b>Others</b>             |         |       |       |       |
| Cash dividend payout (%)  | 66%     | 66%   | 66%   | 66%   |
| Cash div (NT\$)           | 2.00    | 10.38 | 21.92 | 41.73 |
| Yield (%)                 | 0%      | 1%    | 2%    | 4%    |
| Net Debt/Equity (%)       | -19%    | -37%  | -53%  | -67%  |
| Liabilities/Equity (%)    | 33%     | 36%   | 37%   | 36%   |
| Liabilities/Assets (%)    | 25%     | 27%   | 27%   | 26%   |
| ROAE (%)                  | 4%      | 20%   | 34%   | 49%   |
| ROAA (%)                  | 3%      | 15%   | 25%   | 36%   |
| AR/NR Turnover (days)     | 75      | 74    | 75    | 74    |
| Inventory Turnover (days) | 46      | 46    | 46    | 47    |
| AP/NP Turnover (days)     | 16      | 16    | 16    | 16    |
| Cash conversion cycle     | 105     | 105   | 105   | 105   |

Source: Company data, Morgan Stanley Research (E) estimates.

# ZDT: Estimate Changes

We raise our CY2026, 2027 and 2028 EPS estimates by 4% each, reflecting increased demand, pricing, and margin assumptions on ABF substrates.

## Exhibit 30: Estimate Changes

| NT\$mnn              | 2026E    |          |          | 2027E    |          |          | 2028E    |          |          |
|----------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
|                      | New      | Old      | Variance | New      | Old      | Variance | New      | Old      | Variance |
| Net sales            | 226,233  | 226,287  | 0%       | 291,509  | 291,591  | 0%       | 337,259  | 337,349  | 0%       |
| COGS                 | -170,740 | -170,996 | 0%       | -212,926 | -213,614 | 0%       | -239,242 | -240,162 | 0%       |
| Gross profit         | 55,493   | 55,291   | 0%       | 78,584   | 77,976   | 1%       | 98,017   | 97,187   | 1%       |
| Operating expenses   | -28,404  | -29,376  | -3%      | -33,132  | -34,573  | -4%      | -35,256  | -36,776  | -4%      |
| Operating income     | 27,089   | 25,914   | 5%       | 45,452   | 43,403   | 5%       | 62,761   | 60,411   | 4%       |
| Non-operating income | 1,039    | 1,039    | 0%       | 1,406    | 1,406    | 0%       | 1,406    | 1,406    | 0%       |
| Pre-tax income       | 28,128   | 26,953   | 4%       | 46,858   | 44,809   | 5%       | 64,166   | 61,817   | 4%       |
| Net income           | 16,139   | 15,470   | 4%       | 27,071   | 25,909   | 4%       | 37,047   | 35,713   | 4%       |
| EPS (NT\$)           | 15.08    | 14.46    | 4%       | 25.30    | 24.21    | 4%       | 34.62    | 33.37    | 4%       |
| Margins (%)          |          |          | ppt      |          |          | ppt      |          |          | ppt      |
| Gross margin         | 24.5%    | 24.4%    | 0.1      | 27.0%    | 26.7%    | 0.2      | 29.1%    | 28.8%    | 0.3      |
| Operating margin     | 12.0%    | 11.5%    | 0.5      | 15.6%    | 14.9%    | 0.7      | 18.6%    | 17.9%    | 0.7      |
| Pre-tax margin       | 12.4%    | 11.9%    | 0.5      | 16.1%    | 15.4%    | 0.7      | 19.0%    | 18.3%    | 0.7      |
| Net margin           | 7.1%     | 6.8%     | 0.3      | 9.3%     | 8.9%     | 0.4      | 11.0%    | 10.6%    | 0.4      |

Source: Morgan Stanley Research (E) estimates.

## Price Target and Valuation Methodology

**Our base-case value/price target rises ~4% to NT\$690**, derived from our multistage residual income (RI) model. We use the company's beginning equity plus the present value of all expected residual income of earnings in excess of the cost of capital.

We apply the company's beginning equity plus the present value of all expected residual income of earnings in excess of the cost of capital. The RI is positive when ROAE is above the cost of capital and negative when it is below the cost of capital. We assume a cost of equity of 10%, a medium-term growth rate of 15% and a terminal growth rate of 3% (all unchanged).

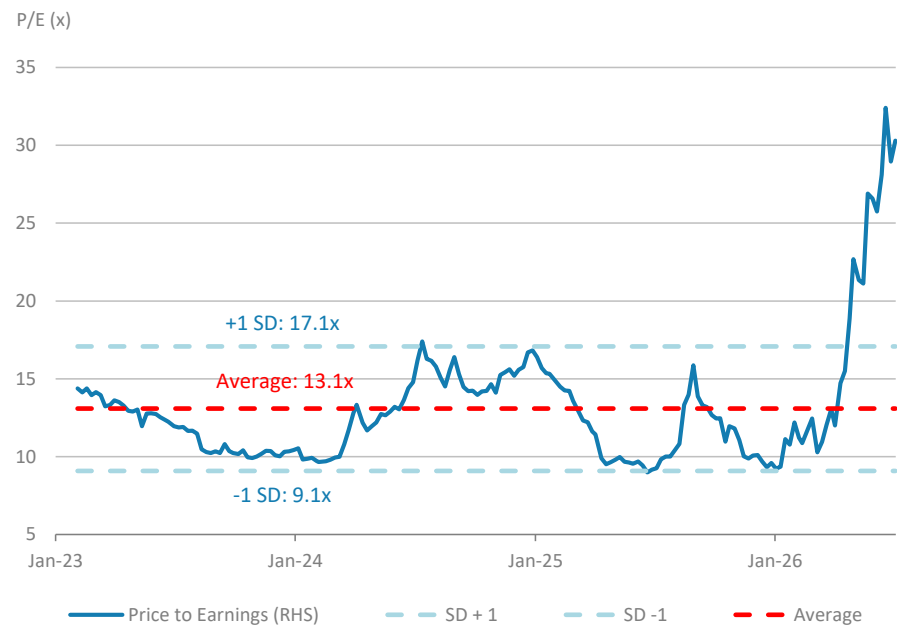
**Our bull- and bear-case values also rises by similar magnitudes to NT\$980 and NT\$400, respectively.**

## Exhibit 31: Residual income (RI) model

| NT\$ mn                     | 2026E   | 2027E   | 2028E   | 2029E   | 2030E   | 2031E   | 2032E   | 2033E   | 2034E   | 2035E   | 2036E   |
|-----------------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| Total Equity                | 183,212 | 203,029 | 227,909 | 239,286 | 252,370 | 267,416 | 284,719 | 304,618 | 327,501 | 353,817 | 384,080 |
| Core Net Profit             | 16,139  | 27,071  | 37,047  | 42,604  | 48,995  | 56,344  | 64,796  | 74,515  | 85,692  | 98,546  | 113,328 |
| Return on Equity            | 9.5%    | 14.8%   | 18.2%   | 18.7%   | 20.5%   | 22.3%   | 24.2%   | 26.2%   | 28.1%   | 30.1%   | 32.0%   |
| Beta (Last 60 Mths)         | 1.20    |         |         |         |         |         |         |         |         |         |         |
| Equity Risk Premium (Rm-Rf) | 6%      |         |         |         |         |         |         |         |         |         |         |
| Risk Free Rate (Rf)         | 3%      |         |         |         |         |         |         |         |         |         |         |
| Cost of Equity              | 10%     |         |         |         |         |         |         |         |         |         |         |
| Terminal Growth Rate        | 3%      |         |         |         |         |         |         |         |         |         |         |
| Continuing Value Spread     | -2%     |         |         |         |         |         |         |         |         |         |         |
| Medium-term growth rate     | 15.0%   |         |         |         |         |         |         |         |         |         |         |
| Residual Income             | -420    | 9,300   | 17,353  | 20,497  | 25,784  | 31,864  | 38,856  | 46,897  | 56,145  | 66,779  | 79,008  |
| Spread                      | 0%      | 5%      | 9%      | 9%      | 11%     | 13%     | 15%     | 16%     | 18%     | 20%     | 22%     |
| Beginning Equity Capital    | 183,212 |         |         |         |         |         |         |         |         |         |         |
| PV of Forecast Period       | 174,680 |         |         |         |         |         |         |         |         |         |         |
| PV of Continuing Value      | 380,487 |         |         |         |         |         |         |         |         |         |         |
| Equity Value                | 738,380 |         |         |         |         |         |         |         |         |         |         |
| No. of Shares               | 1,070   |         |         |         |         |         |         |         |         |         |         |
| Projected Price (EoY)       | 690.0   |         |         |         |         |         |         |         |         |         |         |

Source: Morgan Stanley Research (E) estimates.

Exhibit 32: ZDT P/E chart



Source: TEJ, Morgan Stanley Research estimates.

Financials



## Exhibit 33: Quarterly Estimates

| NT\$ mn                  | 1Q25          | 2Q25          | 3Q25          | 4Q25          | 1Q26          | 2Q26E         | 3Q26E         | 4Q26E         | 2025           | 2026E          | 2027E          | 2028E          |
|--------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|----------------|----------------|----------------|----------------|
| <b>Sales</b>             | <b>40,082</b> | <b>38,203</b> | <b>47,366</b> | <b>56,870</b> | <b>40,728</b> | <b>48,160</b> | <b>64,761</b> | <b>72,583</b> | <b>182,522</b> | <b>226,233</b> | <b>291,509</b> | <b>337,259</b> |
| COGS                     | -34,197       | -31,193       | -36,956       | -44,040       | -31,917       | -37,187       | -48,092       | -53,543       | -146,385       | -170,740       | -212,926       | -239,242       |
| <b>Gross profit</b>      | <b>5,885</b>  | <b>7,011</b>  | <b>10,410</b> | <b>12,830</b> | <b>8,812</b>  | <b>10,972</b> | <b>16,669</b> | <b>19,040</b> | <b>36,136</b>  | <b>55,493</b>  | <b>78,584</b>  | <b>98,017</b>  |
| Operating expenses       | -4,829        | -4,586        | -5,880        | -6,909        | -6,308        | -6,875        | -7,489        | -7,732        | -22,205        | -28,404        | -33,132        | -35,256        |
| - Promotion              | -486          | -493          | -630          | -744          | -616          | -692          | -837          | -845          | -2,352         | -2,990         | -3,780         | -4,151         |
| - ADM                    | -1,781        | -1,483        | -2,270        | -2,512        | -2,116        | -2,378        | -2,558        | -2,666        | -8,046         | -9,718         | -11,737        | -12,890        |
| - R&D                    | -2,563        | -2,611        | -2,981        | -3,653        | -3,576        | -3,806        | -4,094        | -4,221        | -11,807        | -15,697        | -17,615        | -18,215        |
| <b>Operating profit</b>  | <b>1,056</b>  | <b>2,425</b>  | <b>4,530</b>  | <b>5,921</b>  | <b>2,503</b>  | <b>4,097</b>  | <b>9,180</b>  | <b>11,308</b> | <b>13,932</b>  | <b>27,089</b>  | <b>45,452</b>  | <b>62,761</b>  |
| Non-operating income     | 401           | -153          | 131           | -248          | -103          | 314           | 414           | 414           | 131            | 1,039          | 1,406          | 1,406          |
| Interest income          | 119           | 91            | 49            | 95            | 64            | 64            | 64            | 64            | 355            | 256            | 256            | 256            |
| Investment income        | -97           | 203           | 242           | 53            | 431           | 100           | 200           | 200           | 401            | 931            | 550            | 550            |
| Disposal of investment   | 0             | 0             | 0             | 0             | 0             | 0             | 0             | 0             | 0              | 0              | 0              | 0              |
| Disposal of fixed assets | 16            | 58            | -258          | -102          | 0             | 0             | 0             | 0             | -286           | 0              | 0              | 0              |
| Exchange gain            | 103           | -103          | 0             | 0             | 0             | 0             | 0             | 0             | 0              | 0              | 0              | 0              |
| Other                    | 259           | -401          | 98            | -294          | -598          | 150           | 150           | 150           | -338           | -148           | 600            | 600            |
| Pre-tax profit           | 1,457         | 2,272         | 4,661         | 5,673         | 2,400         | 4,411         | 9,594         | 11,722        | 14,063         | 28,128         | 46,858         | 64,166         |
| Income tax               | -431          | -885          | -1,072        | -1,070        | -353          | -970          | -1,804        | -1,407        | -3,458         | -4,534         | -7,282         | -9,992         |
| <b>Net profit</b>        | <b>632</b>    | <b>605</b>    | <b>2,392</b>  | <b>3,161</b>  | <b>1,426</b>  | <b>2,242</b>  | <b>5,183</b>  | <b>7,289</b>  | <b>6,791</b>   | <b>16,139</b>  | <b>27,071</b>  | <b>37,047</b>  |
| <b>EPS (NT\$)</b>        | <b>0.66</b>   | <b>0.63</b>   | <b>2.46</b>   | <b>3.22</b>   | <b>1.33</b>   | <b>2.09</b>   | <b>4.84</b>   | <b>6.81</b>   | <b>6.91</b>    | <b>15.08</b>   | <b>25.30</b>   | <b>34.62</b>   |
| <b>Margins</b>           |               |               |               |               |               |               |               |               |                |                |                |                |
| Gross margin             | 15%           | 18%           | 22%           | 23%           | 22%           | 23%           | 26%           | 26%           | 20%            | 25%            | 27%            | 29%            |
| Operating margin         | 3%            | 6%            | 10%           | 10%           | 6%            | 9%            | 14%           | 16%           | 8%             | 12%            | 16%            | 19%            |
| Pre-tax margin           | 4%            | 6%            | 10%           | 10%           | 6%            | 9%            | 15%           | 16%           | 8%             | 12%            | 16%            | 19%            |
| Net margin               | 2%            | 2%            | 5%            | 6%            | 4%            | 5%            | 8%            | 10%           | 4%             | 7%             | 9%             | 11%            |
| <b>QoQ Growth</b>        |               |               |               |               |               |               |               |               |                |                |                |                |
| Sales                    | -29%          | -5%           | 24%           | 20%           | -28%          | 18%           | 34%           | 12%           |                |                |                |                |
| Gross profit             | -49%          | 19%           | 48%           | 23%           | -31%          | 25%           | 52%           | 14%           |                |                |                |                |
| Operating profit         | -81%          | 130%          | 87%           | 31%           | -58%          | 64%           | 124%          | 23%           |                |                |                |                |
| Pre-tax profit           | -81%          | 56%           | 105%          | 22%           | -58%          | 84%           | 117%          | 22%           |                |                |                |                |
| Net profit               | -86%          | -4%           | 295%          | 32%           | -55%          | 57%           | 131%          | 41%           |                |                |                |                |
| <b>YoY Growth</b>        |               |               |               |               |               |               |               |               |                |                |                |                |
| Sales                    | 23%           | 18%           | -6%           | 1%            | 2%            | 26%           | 37%           | 28%           | 6%             | 24%            | 29%            | 16%            |
| Gross profit             | 10%           | 65%           | -9%           | 12%           | 50%           | 57%           | 60%           | 48%           | 11%            | 54%            | 42%            | 25%            |
| Operating profit         | 42%           | -477%         | -24%          | 7%            | 137%          | 69%           | 103%          | 91%           | 20%            | 94%            | 68%            | 38%            |
| Pre-tax profit           | -3%           | 346%          | -12%          | -27%          | 65%           | 94%           | 106%          | 107%          | -7%            | 100%           | 67%            | 37%            |
| Net profit               | -35%          | 25%           | -29%          | -28%          | 125%          | 271%          | 117%          | 131%          | -26%           | 138%           | 68%            | 37%            |

Source: Company data, Morgan Stanley Research (E) estimates.

**Exhibit 34: Consolidated Financial Summary****Income Statement**

| NT\$ mn                  | 2025           | 2026E          | 2027E          | 2028E          |
|--------------------------|----------------|----------------|----------------|----------------|
| <b>Net sales</b>         | <b>182,522</b> | <b>226,233</b> | <b>291,509</b> | <b>337,259</b> |
| COGS                     | -146,385       | -170,740       | -212,926       | -239,242       |
| <b>Gross profit</b>      | <b>36,136</b>  | <b>55,493</b>  | <b>78,584</b>  | <b>98,017</b>  |
| Operating expenses       | -22,205        | -28,404        | -33,132        | -35,256        |
| - Promotion              | -2,352         | -2,990         | -3,780         | -4,151         |
| - ADM                    | -8,046         | -9,718         | -11,737        | -12,890        |
| - R&D                    | -11,807        | -15,697        | -17,615        | -18,215        |
| <b>Operating income</b>  | <b>13,932</b>  | <b>27,089</b>  | <b>45,452</b>  | <b>62,761</b>  |
| Non-operating income     | 131            | 1,039          | 1,406          | 1,406          |
| Interest income          | 355            | 256            | 256            | 256            |
| Investment income        | 401            | 931            | 550            | 550            |
| Disposal of investment   | 0              | 0              | 0              | 0              |
| Disposal of fixed assets | -286           | 0              | 0              | 0              |
| Exchange gain            | 0              | 0              | 0              | 0              |
| Other                    | -338           | -148           | 600            | 600            |
| Pre-tax income           | 14,063         | 28,128         | 46,858         | 64,166         |
| Income tax               | -7,272         | -11,989        | -19,786        | -27,119        |
| <b>Net income</b>        | <b>6,791</b>   | <b>16,139</b>  | <b>27,071</b>  | <b>37,047</b>  |
| <b>EPS (NT\$)</b>        | <b>6.91</b>    | <b>15.08</b>   | <b>25.30</b>   | <b>34.62</b>   |

**Balance Sheet**

| NT\$ mn                             | 2025           | 2026E          | 2027E          | 2028E          |
|-------------------------------------|----------------|----------------|----------------|----------------|
| Cash                                | 71,116         | 60,966         | 72,632         | 88,446         |
| Mkt securities                      | 36             | 36             | 36             | 36             |
| Accounts/Notes receivables          | 31,412         | 38,935         | 50,169         | 58,043         |
| Inventory                           | 19,616         | 22,880         | 28,533         | 32,060         |
| Other current assets                | 5,460          | 5,460          | 5,460          | 5,460          |
| Current Assets                      | 127,641        | 128,277        | 156,831        | 184,045        |
| Long-term investments               | 9,866          | 10,797         | 11,347         | 11,897         |
| Fixed assets                        | 123,737        | 145,693        | 148,605        | 152,383        |
| Other assets                        | 18,799         | 18,799         | 18,799         | 18,799         |
| <b>Total Assets</b>                 | <b>280,043</b> | <b>303,566</b> | <b>335,582</b> | <b>367,124</b> |
| S/T borrowings                      | 23,839         | 23,839         | 23,839         | 23,839         |
| AP/NP                               | 23,501         | 27,411         | 34,184         | 38,409         |
| Other ST liabilities                | 25,481         | 32,595         | 38,020         | 40,458         |
| Other liabilities                   | 21,151         | 21,151         | 21,151         | 21,151         |
| L/T debt                            | 15,359         | 15,359         | 15,359         | 15,359         |
| <b>Total Liabilities</b>            | <b>109,330</b> | <b>120,354</b> | <b>132,552</b> | <b>139,215</b> |
| Common shares                       | 10,706         | 10,706         | 10,706         | 10,706         |
| Capital Collection                  | 0              | 0              | 0              | 0              |
| APIC                                | 52,902         | 52,902         | 52,902         | 52,902         |
| Retained earnings                   | 61,549         | 73,996         | 93,814         | 118,694        |
| Treasury Stock                      | -52            | 0              | 0              | 0              |
| Other shareholders' equity          | 45607          | 45607          | 45607          | 45607          |
| <b>Shareholders' equity</b>         | <b>170,713</b> | <b>183,212</b> | <b>203,029</b> | <b>227,909</b> |
| <b>Total Liab./Shrhldr's Equity</b> | <b>280,043</b> | <b>303,566</b> | <b>335,582</b> | <b>367,124</b> |

Source: Company data, Morgan Stanley Research (E) estimates.

**Cash Flow Statement**

| NT\$ mn                            | 2025           | 2026E          | 2027E          | 2028E          |
|------------------------------------|----------------|----------------|----------------|----------------|
| <b>Cashflow from operations</b>    | <b>27,617</b>  | <b>23,490</b>  | <b>28,920</b>  | <b>37,981</b>  |
| Net Profits                        | 10,605         | 16,139         | 27,071         | 37,047         |
| Depreciation                       | 18,552         | 8,044          | 7,088          | 6,222          |
| Equity investment losses (income)  | 0              | -931           | -550           | -550           |
| Other adjustments                  | -1,541         | 238            | -4,689         | -4,738         |
| <b>Cashflow from investing</b>     | <b>-31,960</b> | <b>-30,000</b> | <b>-10,000</b> | <b>-10,000</b> |
| (Purchases) sale of FA (capex)     | -32,673        | -30,000        | -10,000        | -10,000        |
| (Purchases) sale of L/T investment | 342            | 0              | 0              | 0              |
| (Purchases) sale of S/T investment | -601           | 0              | 0              | 0              |
| Other adjustments                  | 972            | 0              | 0              | 0              |
| <b>Cashflow from financing</b>     | <b>-1,865</b>  | <b>-3,640</b>  | <b>-7,254</b>  | <b>-12,167</b> |
| Increase in L/T debt               | 6,344          | 0              | 0              | 0              |
| Increase in S/T debt               | 2,191          | 0              | 0              | 0              |
| Issuance of stock                  | 730            | 0              | 0              | 0              |
| Cash dividends                     | -7,221         | -3,692         | -7,254         | -12,167        |
| Dir. & Emp. Bonus                  | 0              | 0              | 0              | 0              |
| Changes in Treasury Stocks         | 288            | 52             | 0              | 0              |
| Other adjustments                  | -4,196         | 0              | 0              | 0              |
| Exchange rate adjustment           | -2,177         | 0              | 0              | 0              |
| <b>Net change in cash</b>          | <b>-8,386</b>  | <b>-10,150</b> | <b>11,666</b>  | <b>15,814</b>  |

**Financial Ratios**

|                                    | 2025   | 2026E  | 2027E | 2028E |
|------------------------------------|--------|--------|-------|-------|
| <b>Margins</b>                     |        |        |       |       |
| Gross margin                       | 19.8%  | 24.5%  | 27.0% | 29.1% |
| Operating margin                   | 7.6%   | 12.0%  | 15.6% | 18.6% |
| Pretax margin                      | 7.7%   | 12.4%  | 16.1% | 19.0% |
| Net margin                         | 3.7%   | 7.1%   | 9.3%  | 11.0% |
| <b>YoY growth</b>                  |        |        |       |       |
| Sales                              | 6.3%   | 23.9%  | 28.9% | 15.7% |
| Operating profits                  | 20.2%  | 94.4%  | 67.8% | 38.1% |
| Pretax profits                     | -6.5%  | 100.0% | 66.6% | 36.9% |
| Net profits                        | -26.0% | 137.7% | 67.7% | 36.8% |
| EPS                                | -28.6% | 118.3% | 67.7% | 36.8% |
| Net Debt/Equity (Net of mkt secs.) | -19%   | -12%   | -16%  | -22%  |
| Net Debt/Equity                    | -19%   | -12%   | -16%  | -22%  |
| Liabilities/Equity                 | 64%    | 66%    | 65%   | 61%   |
| Liabilities/Assets                 | 39%    | 40%    | 39%   | 38%   |
| ROAE                               | 4.2%   | 9.1%   | 14.0% | 17.2% |
| ROAA                               | 2.5%   | 5.5%   | 8.5%  | 10.5% |
| AR/NR Turnover (days)              | 62     | 57     | 56    | 59    |
| AP/NP Turnover (days)              | 56     | 54     | 53    | 55    |
| Inventory Turnover (days)          | 47     | 45     | 44    | 46    |
| Cash conversion cycle (days)       | 52     | 48     | 47    | 49    |

Risk Reward – Unimicron (3037.TW)

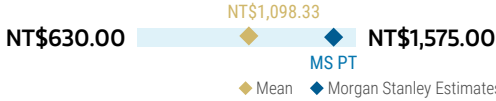
Price hikes and spec upgrades to support higher earnings estimates; OW

PRICE TARGET NT\$1,465.00

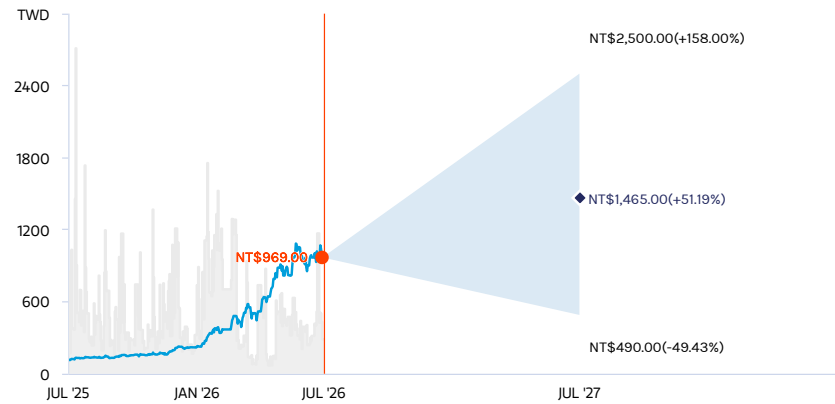
Base case, residual income (RI) valuation model, which we think derives the most accurate value of the firm given that it takes into account cost of equity. We use a cost of equity of 9.2% [risk-free rate of 1% (10-year Taiwan government note yield), equity risk premium of 8.7%, and a beta of 1.0], a medium-term growth rate of 15%, and a terminal growth rate of 3%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



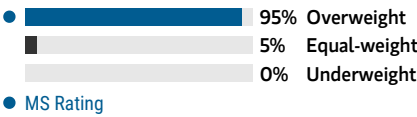
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We expect the ABF substrate market to revert to undersupply from 2027 onward, with the undersupply gap reaching 20-25% by 2030.
- This is primarily driven by AI demand, with next-gen AI/networking chips adopting larger body size ABF substrates.
- In the near to medium term, the short supply of T-glass will be a key watch point, but we believe this will mainly affect low-end ABF and BT substrates, as suppliers prioritize supply to the AI chip vendors.
- Our price target implies 44x 2027e P/E or 25x 2028e P/E, above the 10-20x range at which it traded in the previous upcycle. We deem this fair, given AI chip substrate spec upgrades and suppliers' ability to raise prices.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: Positive  
Secular Growth: Positive

View descriptions of Risk Rewards Themes [here](#)

**BULL CASE** NT\$2,500.00  
75x 2027e P/E

**Demand for AI servers, general servers and PCs is much stronger than expected:** Pricing increases more than expected, 30%+ in 2026-27, driving stronger margin expansion for Unimicron.

**BASE CASE** NT\$1,465.00  
44x 2027e P/E

**Entering up-cycle driven by AI chip upgrades:** Robust demand for AI GPUs and ASICs, supported by continued data center infrastructure investments by CSPs and neo cloud, leads to undersupply in the high-end ABF market from 2027 onward. PC demand is sluggish, while general servers still show double-digit unit growth in 2026. ASP has bottomed and started to inflect upward, driven by raw material price hikes and substrate spec upgrades.

**BEAR CASE** NT\$490.00  
15x 2027e P/E

**Demand for AI servers and general servers softens, with PC demand weakening further:** The result is greater ABF oversupply; thus, there are pricing declines of ~15-20% in 2026-27, leading to weaker margins for Unimicron over the next two years.

Risk Reward – Unimicron (3037.TW)

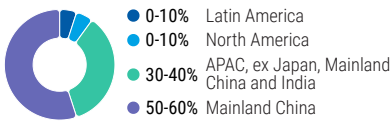
KEY EARNINGS INPUTS

| Drivers                  | Dec 2025 | Dec 2026e | Dec 2027e | Dec 2028e |
|--------------------------|----------|-----------|-----------|-----------|
| IC substrate ASP YoY (%) | 7.5      | 30.0      | 35.1      | 28.9      |
| IC substrate GM (%)      | 15.2     | 27.3      | 36.6      | 42.8      |

INVESTMENT DRIVERS

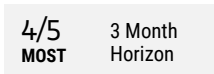
- 5G smartphone demand
- 5G base station demand
- Automotive demand

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate  
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS



Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Better-than-expected ABF substrate demand from PC and server customers
- Capex cuts; halt to its capacity expansion plan
- Continued yield issues of alternative technology that doesn't require substrate (e.g., CoWoP)

RISKS TO DOWNSIDE

- Sudden demand shortfall
- Technological change that would not require ABF substrates
- Intensifying competition
- Yield issues or production hiccups when ramping new capacity

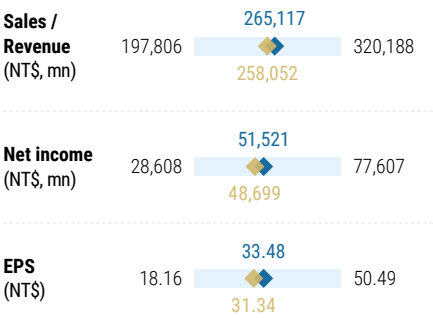
OWNERSHIP POSITIONING



Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e



Source: Refinitiv, Morgan Stanley Research

Risk Reward – Nan Ya PCB (8046.TW)

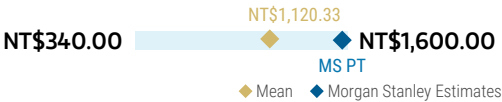
Price hikes and spec upgrades to support higher earnings estimates; OW

PRICE TARGET NT\$1,550.00

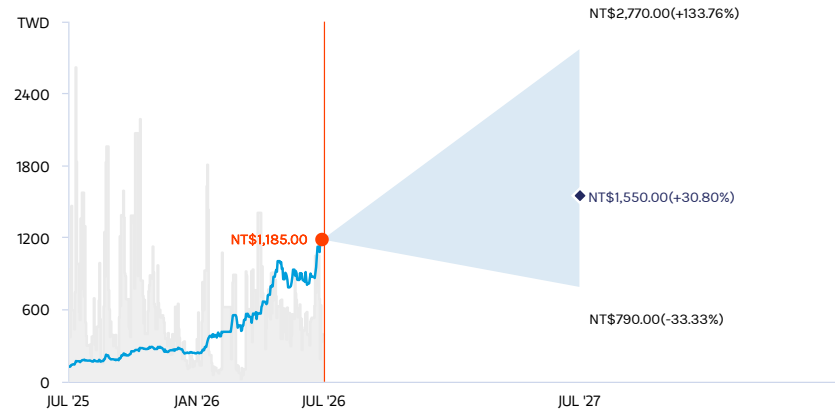
Base case, residual income (RI) valuation model, which we think derives the most accurate value of the firm given that it takes into account cost of equity. We use a cost of equity of 9.3% [risk-free rate of 1% (10-year Taiwan government note yield), equity risk premium of 8.7%, and a beta of 1.0], a medium-term growth rate of 17%, and a terminal growth rate of 3%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



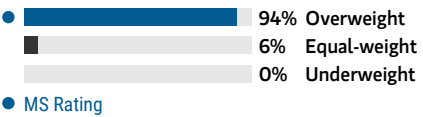
Key: — Historical Stock Performance      ● Current Stock Price      ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- Price hikes for BT and ABF substrates, driven by raw material cost increases, will likely drive margin improvement for NYPCB.
- We believe the ABF substrate market will become undersupplied again from 2027 onward, driven by AI demand, with next-gen AI/networking chips adopting large-body size ABF substrates.
- NYPCB should benefit from ABF substrate structural tailwinds as well - rising data traffic, big data analysis, AI/ML, and 5G.
- Our price target implies 47x 2027e P/E or 25x 2028e P/E, vs. the 10-30x range in which it traded in the previous upcycle. We deem this fair given ABF substrate spec upgrades and suppliers' ability to raise prices.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: Positive

Secular Growth: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

NT\$2,770.00

84x 2027e P/E

BT price increases further in 1H26, and ABF becomes undersupplied faster than expected (current expectation is from 2027 onward), driving even stronger margin expansion for Nan Ya PCB.

BASE CASE

NT\$1,550.00

47x 2027e P/E

**Entering up-cycle driven by AI chip upgrades:** Robust demand for AI GPUs/ASICs and networking chips, supported by continued data center infrastructure investments by CSPs and neo cloud, leads to undersupply in the high-end ABF market from 2027 onward. PC demand is sluggish, while general servers still show double-digit unit growth in 2026. ASP has bottomed and started to inflect upward, driven by raw material price hikes and substrate spec upgrades.

BEAR CASE

NT\$790.00

24x 2027e P/E

ABF demand weakens further, resulting in greater oversupply in the next few years. Thus, ABF pricing decreases 20% or more in 2026-27, leading to weaker margins for Nan Ya PCB.

Risk Reward – Nan Ya PCB (8046.TW)

KEY EARNINGS INPUTS

| Drivers         | Dec 2025 | Dec 2026e | Dec 2027e | Dec 2028e |
|-----------------|----------|-----------|-----------|-----------|
| ABF ASP y/y (%) | 10.0     | 20.0      | 33.9      | 38.6      |
| ABF GM (%)      | 17.3     | 30.5      | 41.4      | 53.0      |

INVESTMENT DRIVERS

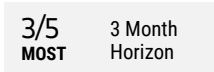
- 5G smartphone demand
- 5G base station demand
- Automotive demand

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate  
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS



Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Better-than-expected ABF and BT demand
- Faster-than-expected pickup in AI and 5G demand
- Stronger-than-expected ASP hike
- Continued yield issues of alternative technology that doesn't require substrate (e.g. CoWoP)

RISKS TO DOWNSIDE

- Sudden demand shortfall, creating headwinds for ABF substrate demand and pricing
- Technological change that would not require ABF substrates
- Intensifying competition

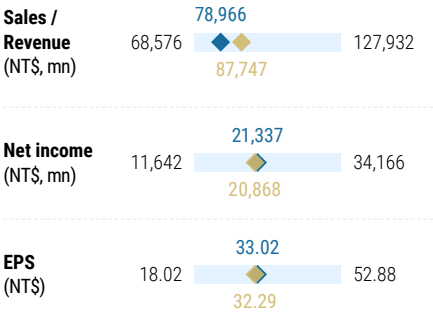
OWNERSHIP POSITIONING



Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Zhen Ding (4958.TW)

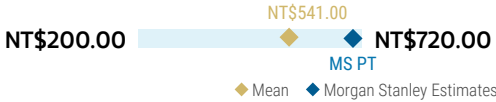
Growing China and international AI chip substrate and PCB vendor

PRICE TARGET NT\$690.00

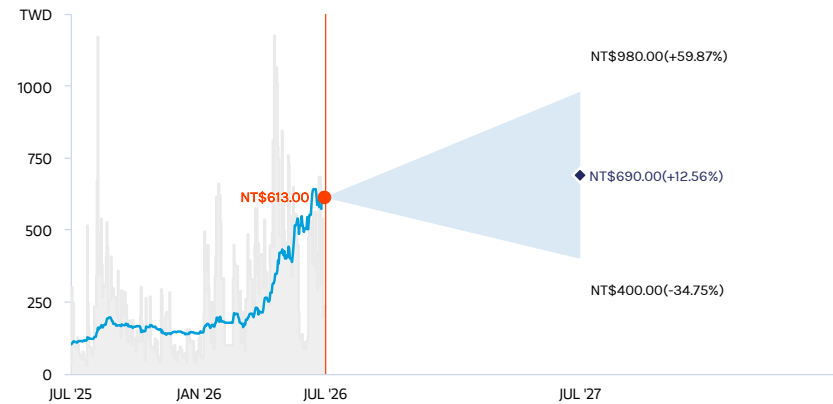
Base case, residual income model. Key assumptions include a cost of equity of 10%, a medium-term growth rate of 15% and a terminal growth rate of 3%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



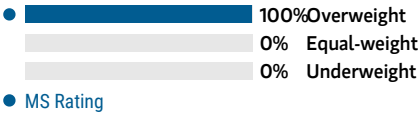
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- iPhone volumes are more resilient than Android given Apple's advantages on securing memory.
- ZDT is one of the key ABF substrate suppliers for China AI chips, which is expected to grow meaningfully over the next few years.
- ZDT is growing its ABF exposure to International clients as well, including Google TPU.
- AI optical transceiver module PCB is a meaningful growth driver for ZDT, with very good margins (GM 40%+).
- Shenzhen ABF fab 1 continues to ramp and has turned profitable in 1Q26, and fab 2 is now under construction. Kaohsiung fab is on track to enter mass production in 2H26.
- Our PT implies 27x/20x CY27e/28e P/E.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: Negative

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

NT\$980.00

39x 2027e P/E

**Greater share gains and better earnings contribution from new projects in ABF and AI Server PCBs:** In the bull case, we assume ZDT is able to do Google TPU substrate extremely well and uses it to approach other ASIC/GPU customers, expanding its international client base. Additionally, we also assume ZDT expands Nvidia AI PCB share to 20-30% share on good execution and yield.

BASE CASE

NT\$690.00

27x 2027e P/E

**Stable demand for F-PCB and SLP, with share gains in BT/ABF substrates, and AI server PCBs:** Zhen Ding is one of the main ABF substrate suppliers for China AI chips, while growing its share for international clients, potentially starting with Google TPU. At the same time, it continues to grow its AI server PCB share with Nvidia and other hyperscalers, and grows its exposure to optical module transceivers.

BEAR CASE

NT\$400.00

16x 2027e P/E

**Weaker iPhone demand with share loss and ZDT does not execute on Google TPU substrates:** iPhone demand misses market expectations, Zhen Ding loses share in F-PCB to Chinese peers, and SLP demand is also weak. We also assume here that ZDT fails to execute on TPU substrates and only continues to have exposure to China AI chips, with less than 5% share for Nvidia AI PCBs.

Risk Reward – Zhen Ding (4958.TW)

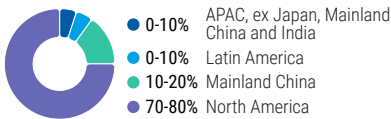
KEY EARNINGS INPUTS

| Drivers                    | Dec 2025 | Dec 2026e | Dec 2027e | Dec 2028e |
|----------------------------|----------|-----------|-----------|-----------|
| F-PCB sales y/y (%)        | 5.0      | 1.2       | 14.6      | 7.3       |
| F-PCB GM (%)               | 21.3     | 24.1      | 24.6      | 25.6      |
| F-PCB ASP per sq ft (NT\$) | 59.5     | 60.4      | 63.1      | 65.3      |

INVESTMENT DRIVERS

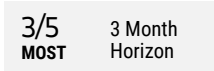
- iPhone, MacBook, iPad demand
- F-PCB content per iPhone, MacBook, iPad
- F-PCB ASP
- Yield rate for ABF/BT substrates

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate  
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS



Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the Least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Higher-than-expected F-PCB content increase for iPhones
- Better-than-expected production yield/share allocation for SLP
- Share gains on higher margin F-PCB pieces

RISKS TO DOWNSIDE

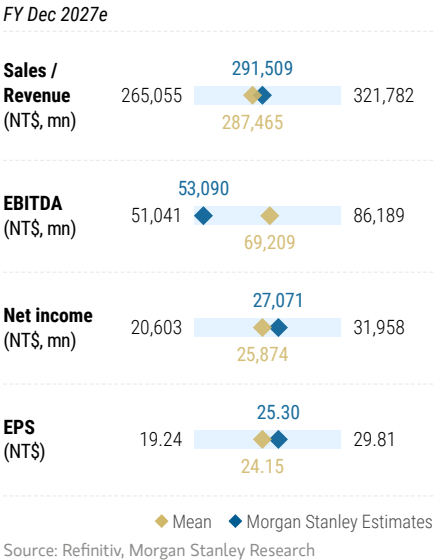
- Worse-than-expected iPhone sell-through
- Lower-than-expected F-PCB content increase for iPhones
- Worse-than-expected production yield/share allocation for SLP
- Increasing competition from Chinese peers, intensifying pricing pressure

OWNERSHIP POSITIONING



Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS





# Valuation Methodology and Risks

## Samsung Electro-Mechanics (009150.KS)

Base case, residual income valuation model. We apply a 10% cost of equity (beta 1.0, equity risk premium of 6.5%, and risk-free rate of 3.5%) and a 5% terminal growth rate. Our base year is 2026 and our forecast period is 2026-36.

### Risks to Upside

- Price hikes driven by MLCC shortage
- Better-than-expected smartphone demand
- Higher MLCC margin from better yield and product mix
- Strong tailwind from China's consumption-friendly policies

### Risks to Downside

- Significant pullback in Samsung's mobile flagship product cycle and SEMCO's lower market share
- Execution risk in penetrating Chinese smartphone customers
- Consumer demand headwinds in 2026

## Ibiden (4062.T)

Derived from the base case. DCF assumptions: 2.6% risk-free rate, 1.40 beta, 3.2% risk premium, for 6.9% WACC. Terminal growth of 0% from F3/36.

### Risks to Upside

- High-value-added ABF package products support earnings increase beyond our expectation from F3/27 2H.
- Sales exceed our base-case scenario as material shortages for FC packages are resolved.

### Risks to Downside

- Sales are weaker than base case because due to intensified competition and declining market share in ABF packages.
- We estimate OP sensitivity to FX changes at ¥800mn for ¥1/US\$ and ¥100mn for ¥1/€.

# Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options\\_Probabilities\\_Exhibit\\_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR\\_Themes\\_Exhibit\\_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG\\_Exhibit\\_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG\\_Sustainable\\_Solutions\\_External\\_Link.pdf](#)
5. View explanation of HERS methodology - [ESG\\_HERS\\_External\\_Link.pdf](#)

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(as of June 30, 2026)

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| Coverage Universe     |       |            | Investment Banking Clients (IBC) |                |                      | Other Material Investment Services Clients (MISC) |                       |
|-----------------------|-------|------------|----------------------------------|----------------|----------------------|---------------------------------------------------|-----------------------|
| Stock Rating Category | Count | % of Total | Count                            | % of Total IBC | % of Rating Category | Count                                             | % of Total Other MISC |
| Overweight/Buy        | 1544  | 42%        | 453                              | 49%            | 29%                  | 757                                               | 44%                   |
| Equal-weight/Hold     | 1577  | 43%        | 390                              | 42%            | 25%                  | 769                                               | 44%                   |
| Not-Rated/Hold        | 3     | 0%         | 1                                | 0%             | 33%                  | 1                                                 | 0%                    |
| Underweight/Sell      | 544   | 15%        | 89                               | 10%            | 16%                  | 204                                               | 12%                   |
| Total                 | 3,668 |            | 933                              |                |                      | 1731                                              |                       |

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

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Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

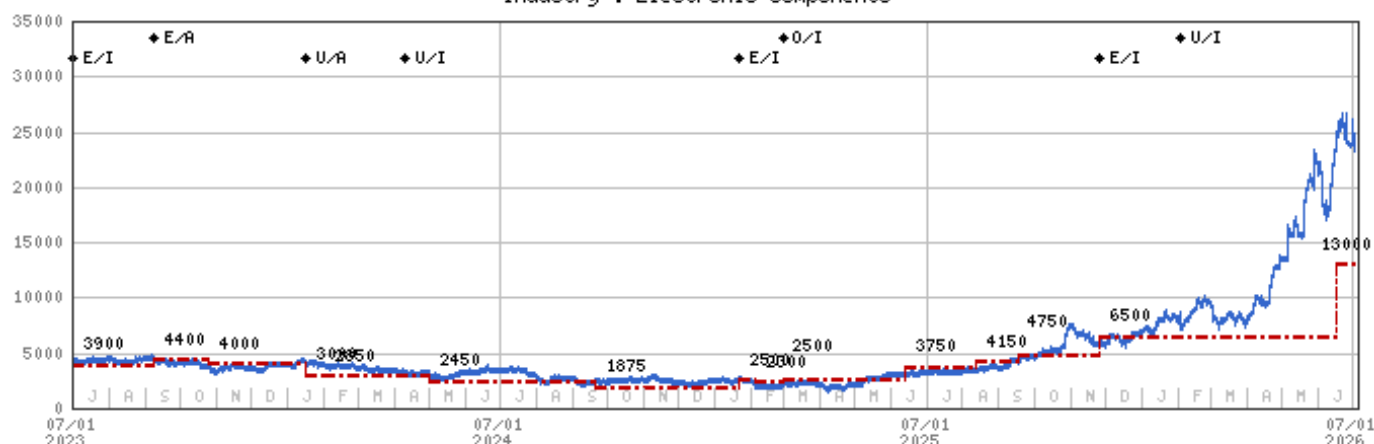
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

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Stock Price, Price Target and Rating History (See Rating Definitions)

Ibiden (4062.T) - As of 07/05/26 GMT in JPY  
Industry : Electronic Components



Stock Rating History: 7/1/21 : O/I; 6/1/22 : U/I; 6/6/23 : E/I; 9/8/23 : E/A; 1/17/24 : U/A; 4/10/24 : U/I; 1/22/25 : E/I;  
2/28/25 : O/I; 11/26/25 : E/I; 2/4/26 : U/I

Price Target History: 6/23/21 : 3650; 8/24/21 : 3750; 9/27/21 : 4000; 10/29/21 : 4200; 1/17/22 : 4250; 2/4/22 : 4100; 4/11/22 : 3750;  
6/1/22 : 1875; 6/30/22 : 1550; 10/27/22 : 1750; 1/17/23 : 2000; 2/3/23 : 1750; 3/22/23 : 1700; 6/6/23 : 3900; 9/8/23 : 4400;  
10/26/23 : 4000; 1/17/24 : 3000; 2/1/24 : 2850; 5/2/24 : 2450; 9/20/24 : 1875; 1/22/25 : 2500; 2/4/25 : 2300; 2/28/25 : 2500;  
6/13/25 : 3750; 8/13/25 : 4150; 9/17/25 : 4750; 11/26/25 : 6500; 6/17/26 : 13000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

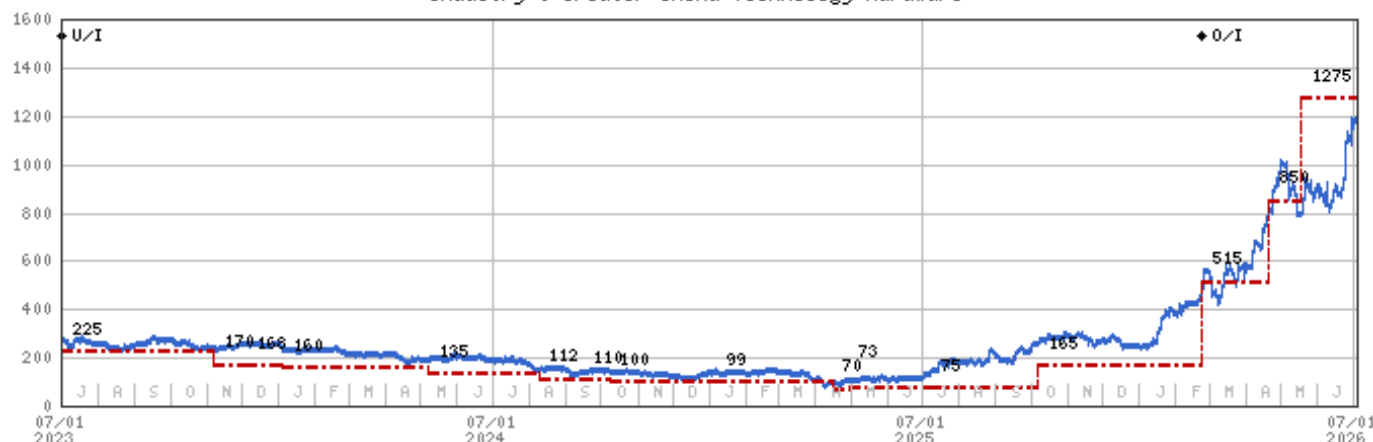
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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Nan Ya PCB (8046.TW) - As of 07/04/26 GMT in TWD  
Industry : Greater China Technology Hardware



Stock Rating History: 7/1/21 : NA/I; 10/22/21 : O/I; 6/1/22 : E/I; 12/21/22 : U/I; 2/23/26 : O/I

Price Target History: 10/22/21 : 580; 11/9/21 : 630; 2/9/22 : 650; 2/25/22 : 670; 6/1/22 : 420; 8/23/22 : 275; 10/27/22 : 235; 12/21/22 : 180; 2/2/23 : 165; 2/25/23 : 155; 6/7/23 : 225; 11/7/23 : 170; 12/4/23 : 168; 1/4/24 : 160; 5/7/24 : 135; 8/9/24 : 112; 9/15/24 : 110; 10/8/24 : 100; 1/6/25 : 99; 4/15/25 : 70; 4/29/25 : 73; 7/7/25 : 75; 10/6/25 : 165; 2/23/26 : 515; 4/20/26 : 850; 5/18/26 : 1275

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

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Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

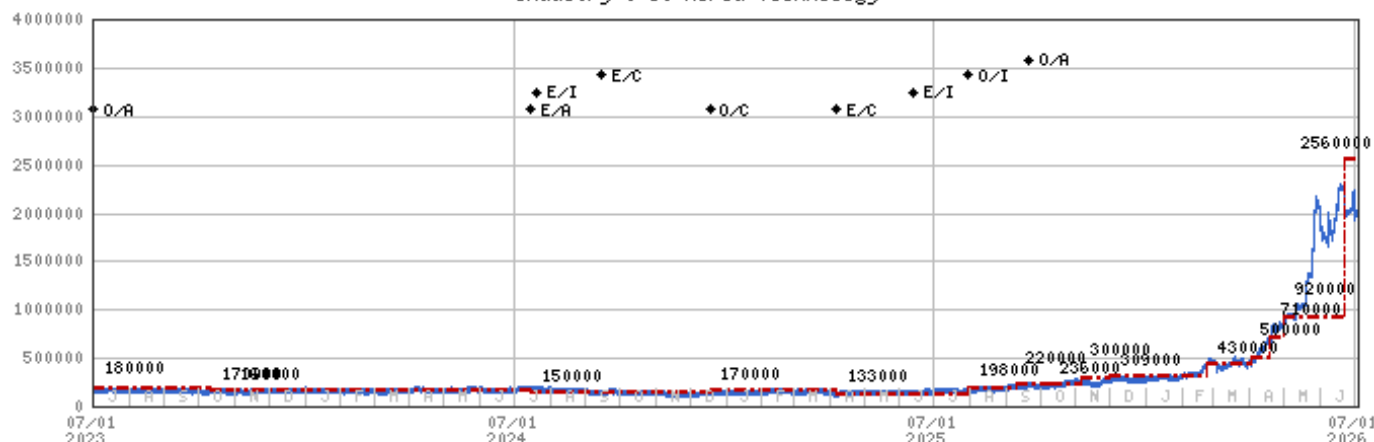
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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Samsung Electro-Mechanics (009150.KS) - As of 07/04/26 GMT in KRW  
Industry : S. Korea Technology



Stock Rating History: 7/1/21 : E/A; 7/19/21 : E/I; 8/12/21 : E/C; 6/1/22 : U/C; 10/4/22 : U/A; 11/9/22 : O/A; 7/15/24 : E/A;  
7/21/24 : E/I; 9/15/24 : E/C; 12/18/24 : O/C; 4/7/25 : E/C; 6/13/25 : E/I; 7/31/25 : O/I; 9/21/25 : O/A

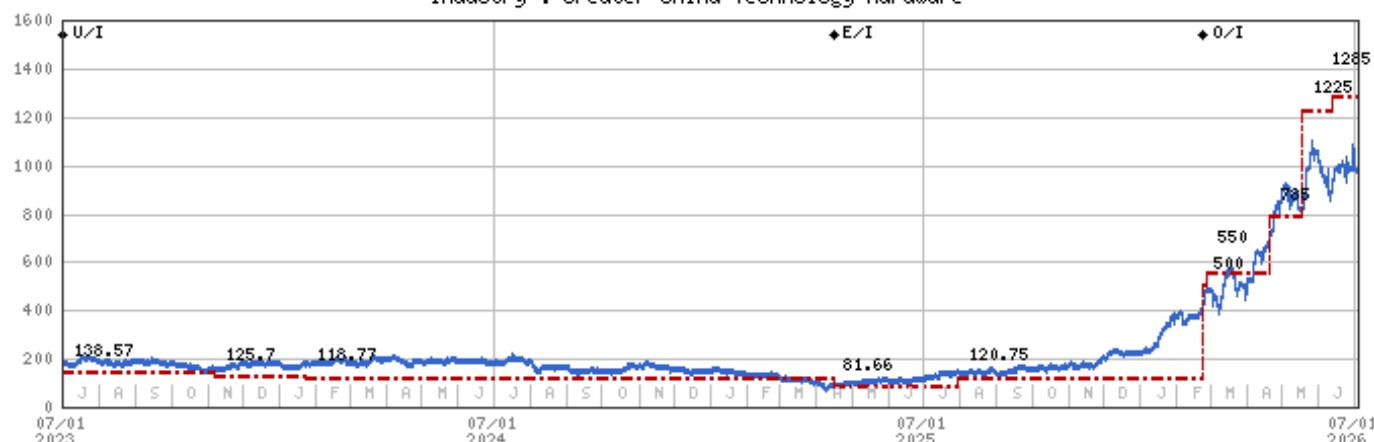
Price Target History: 5/12/21 : 180000; 10/19/21 : 170000; 10/27/21 : 160000; 1/24/22 : 180000; 3/18/22 : 170000; 6/1/22 : 140000;  
11/9/22 : 180000; 10/11/23 : 170000; 10/27/23 : 160000; 7/15/24 : 150000; 12/18/24 : 170000; 4/7/25 : 133000; 7/31/25 : 198000;  
9/10/25 : 220000; 10/10/25 : 236000; 11/5/25 : 300000; 12/1/25 : 309000; 2/23/26 : 430000; 4/2/26 : 500000; 4/19/26 : 710000;  
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Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View  
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)  
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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Unimicron (3037.TW) - As of 07/04/26 GMT in TWD  
Industry : Greater China Technology Hardware



Stock Rating History: 7/1/21 : NA/I; 10/22/21 : O/I; 6/1/22 : E/I; 2/22/23 : U/I; 4/15/25 : E/I; 2/23/26 : O/I

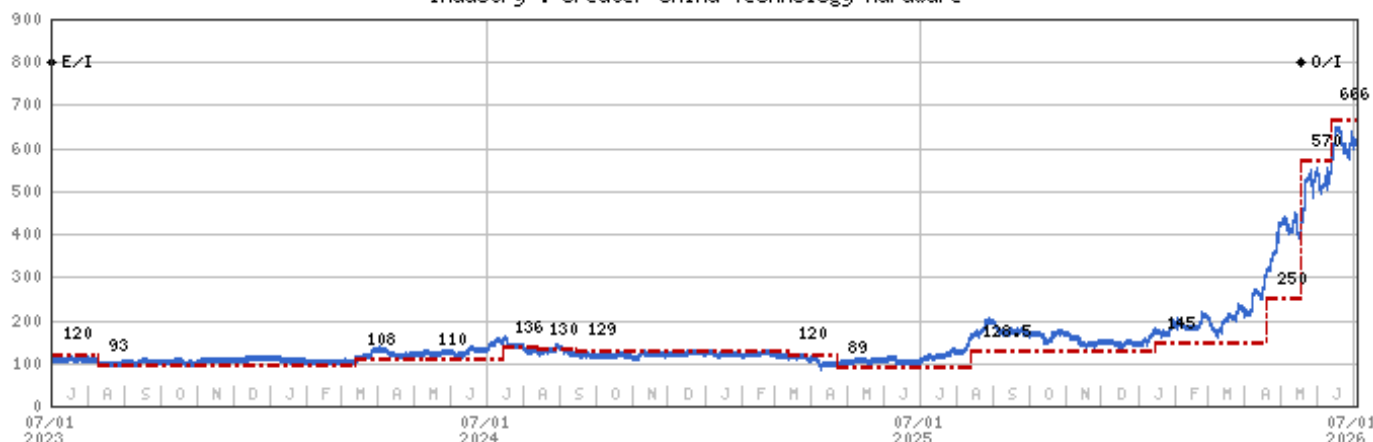
Price Target History: 10/22/21 : 188.06; 10/27/21 : 237.55; 2/9/22 : 267.24; 2/24/22 : 366.22; 6/1/22 : 237.55; 8/23/22 : 178.16;  
10/27/22 : 133.62; 12/21/22 : 123.72; 2/2/23 : 116.79; 2/22/23 : 98.98; 6/7/23 : 138.57; 11/7/23 : 125.7; 1/24/24 : 118.77;  
4/15/25 : 81.66; 7/30/25 : 120.75; 2/23/26 : 500; 2/26/26 : 550; 4/20/26 : 785; 5/18/26 : 1225; 6/12/26 : 1285

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Stock Price (Not Covered by Current Analyst) --- Stock Price (Covered by Current Analyst) ---  
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View  
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)  
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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Zhen Ding (4958.TW) - As of 07/04/26 GMT in TWD  
Industry : Greater China Technology Hardware



Stock Rating History: 7/1/21 : U/I; 8/2/22 : E/I; 5/18/26 : O/I

Price Target History: 3/30/21 : 105; 8/2/22 : 124; 11/4/22 : 126; 3/14/23 : 120; 8/10/23 : 93; 3/13/24 : 108; 5/11/24 : 110; 7/15/24 : 136; 8/13/24 : 130; 9/15/24 : 129; 3/12/25 : 120; 4/23/25 : 89; 8/13/25 : 128.5; 1/16/26 : 145; 4/20/26 : 250; 5/18/26 : 570; 6/12/26 : 666

Source: Morgan Stanley Research      Date Format : MM/DD/YY      Price Target ---      No Price Target Assigned (NA)  
Stock Price (Not Covered by Current Analyst) --- Stock Price (Covered by Current Analyst) ---  
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View  
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)  
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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## INDUSTRY COVERAGE: Greater China Technology Hardware

| COMPANY (TICKER)                                     | RATING (AS OF) | PRICE* (07/03/2026) |
|------------------------------------------------------|----------------|---------------------|
| <b>Andy Meng, CFA</b>                                |                |                     |
| AAC Technologies Holdings (2018.HK)                  | O (01/29/2024) | HK\$42.30           |
| Accelink Technologies Co. Ltd. (002281.SZ)           | U (05/12/2022) | Rmb217.33           |
| BYD Electronics (0285.HK)                            | O (04/28/2023) | HK\$22.26           |
| China TransInfo Technology Co Ltd (002373.SZ)        | E (07/18/2023) | Rmb7.20             |
| Dahua Technology Co. Ltd. (002236.SZ)                | E (12/12/2024) | Rmb16.80            |
| Eoptolink Technology Inc Ltd (300502.SZ)             | O (02/27/2026) | Rmb526.00           |
| Genius Electronic Optical Co. Ltd. (3406.TW)         | E (04/23/2025) | NT\$636.00          |
| Gosuncn Technology Group Co Ltd (300098.SZ)          | U (11/07/2022) | Rmb4.55             |
| HIKVision Digital Technology (002415.SZ)             | E (12/12/2024) | Rmb34.00            |
| Largan Precision (3008.TW)                           | E (10/17/2025) | NT\$4,230.00        |
| LianChuang Electronic Technology Co Ltd (002036.SZ)  | U (06/12/2024) | Rmb7.42             |
| OFILM Group Co Ltd (002456.SZ)                       | U (06/12/2024) | Rmb9.26             |
| Q Technology (Group) Company Ltd (1478.HK)           | U (03/23/2026) | HK\$6.99            |
| Quectel Wireless Solutions Co Ltd (603236.SS)        | E (10/09/2024) | Rmb51.80            |
| Shanghai Conant Optical Co Ltd (2276.HK)             | O (04/10/2026) | HK\$36.08           |
| Shenzhen Transsion Holdings Co Ltd (688036.SS)       | E (03/23/2026) | Rmb60.26            |
| Sunny Optical (2382.HK)                              | E (03/23/2026) | HK\$60.05           |
| Suzhou TFC Optical Communication Co Ltd. (300394.SZ) | E (11/06/2025) | Rmb250.10           |
| Wingtech Technology Co Ltd (600745.SS)               | U (05/18/2026) | Rmb18.25            |
| Xiaomi Corp (1810.HK)                                | O (04/14/2021) | HK\$22.96           |
| Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)  | U (10/13/2021) | Rmb475.08           |
| Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)    | E (04/20/2023) | HK\$201.20          |
| Yongxin Optics Co Ltd (603297.SS)                    | E (11/15/2022) | Rmb127.11           |
| Zhejiang Crystal-Optech Co Ltd (002273.SZ)           | O (11/15/2022) | Rmb36.98            |
| Zhongji Innolight Co Ltd (300308.SZ)                 | ++             | Rmb1,116.00         |

|                                                   |                |              |
|---------------------------------------------------|----------------|--------------|
| ZTE Corporation (0763.HK)                         | O (06/04/2026) | HK\$22.84    |
| ZTE Corporation (000063.SZ)                       | E (06/04/2026) | Rmb36.38     |
| <b>Derrick Yang</b>                               |                |              |
| Accton Technology Corporation (2345.TW)           | O (06/06/2024) | NT\$2,735.00 |
| Advantech (2395.TW)                               | O (01/20/2021) | NT\$521.00   |
| AirTAC International (1590.TW)                    | O (04/16/2025) | NT\$1,440.00 |
| AU Optronics (2409.TW)                            | E (02/10/2026) | NT\$30.00    |
| Bizlink (3665.TW)                                 | O (03/10/2025) | NT\$1,995.00 |
| BOE Technology (000725.SZ)                        | O (09/06/2019) | Rmb8.38      |
| Chenbro (8210.TW)                                 | O (07/23/2025) | NT\$1,305.00 |
| Chroma Ate Inc. (2360.TW)                         | O (10/05/2021) | NT\$2,265.00 |
| E Ink Holdings Inc. (8069.TWO)                    | O (05/11/2026) | NT\$212.50   |
| Ennostar Inc (3714.TW)                            | U (09/23/2022) | NT\$62.30    |
| Hiwin Technologies Corp. (2049.TW)                | O (03/30/2026) | NT\$350.00   |
| Innolux (3481.TW)                                 | E (04/07/2025) | NT\$66.00    |
| King Slide Works Co. Ltd. (2059.TW)               | O (11/08/2023) | NT\$8,000.00 |
| Lens Technology (300433.SZ)                       | E (07/22/2020) | Rmb52.42     |
| Radiant Opto-Electronics Corporation (6176.TW)    | E (03/01/2024) | NT\$90.30    |
| Sanan Optoelectronics (600703.SS)                 | U (08/21/2023) | Rmb18.23     |
| TCL Corp. (000100.SZ)                             | E (04/07/2025) | Rmb5.65      |
| Tianma Microelectronics (000050.SZ)               | U (01/24/2018) | Rmb9.05      |
| Wuhan Jingce Electronic Group Co Ltd (300567.SZ)  | E (11/26/2021) | Rmb263.92    |
| <b>Howard Kao</b>                                 |                |              |
| Acer Inc. (2353.TW)                               | U (04/23/2025) | NT\$32.90    |
| Asustek Computer Inc. (2357.TW)                   | U (11/16/2025) | NT\$675.00   |
| Compal Electronics (2324.TW)                      | U (04/23/2025) | NT\$34.95    |
| FIT Hon Teng Ltd (6088.HK)                        | O (11/03/2025) | HK\$6.79     |
| Giga-Byte Technology Co. Ltd. (2376.TW)           | E (11/16/2025) | NT\$338.50   |
| Gold Circuit Electronics Ltd. (2368.TW)           | O (10/06/2022) | NT\$1,305.00 |
| Inspur Electronic Information (000977.SZ)         | E (08/28/2023) | Rmb66.35     |
| Lenovo (0992.HK)                                  | E (11/16/2025) | HK\$21.30    |
| Lotes Co. Ltd. (3533.TW)                          | E (05/12/2025) | NT\$2,175.00 |
| Nan Ya PCB (8046.TW)                              | O (02/23/2026) | NT\$1,185.00 |
| Pegatron Corporation (4938.TW)                    | E (03/25/2026) | NT\$83.60    |
| Quanta Computer Inc. (2382.TW)                    | O (05/01/2023) | NT\$377.00   |
| Shengyi Technology Co Ltd. (600183.SS)            | E (05/26/2022) | Rmb160.78    |
| Shennan Circuits Co Ltd (002916.SZ)               | E (08/24/2023) | Rmb454.48    |
| Unimicron (3037.TW)                               | O (02/23/2026) | NT\$969.00   |
| Wistron Corporation (3231.TW)                     | O (07/12/2023) | NT\$159.00   |
| Wiwynn Corp (6669.TW)                             | O (11/10/2025) | NT\$5,265.00 |
| Yageo Corp. (2327.TW)                             | O (10/28/2025) | NT\$1,045.00 |
| Zhen Ding (4958.TW)                               | O (05/18/2026) | NT\$613.00   |
| <b>Sharon Shih</b>                                |                |              |
| Asia Vital Components Co. Ltd. (3017.TW)          | O (07/30/2024) | NT\$2,760.00 |
| Auras Technology Co Ltd (3324.TWO)                | E (05/04/2023) | NT\$1,020.00 |
| Catcher Technology (2474.TW)                      | U (11/17/2025) | NT\$207.00   |
| Delta Electronics Inc. (2308.TW)                  | O (07/13/2017) | NT\$2,075.00 |
| Fositek Corp (6805.TW)                            | O (06/25/2025) | NT\$1,630.00 |
| Foxconn Industrial Internet Co. Ltd. (601138.SS)  | O (07/10/2019) | Rmb64.72     |
| GoerTek Inc (002241.SZ)                           | U (04/23/2025) | Rmb22.05     |
| Hon Hai Precision (2317.TW)                       | O (03/15/2024) | NT\$240.50   |
| LandMark Optoelectronics Corporation (3081.TWO)   | E (03/26/2026) | NT\$2,100.00 |
| Lingyi Itech Guangdong Co (002600.SZ)             | U (04/23/2025) | Rmb18.02     |
| Lite-On Technology (2301.TW)                      | E (01/15/2025) | NT\$219.00   |
| Luxshare Precision Industry Co., Ltd. (002475.SZ) | O (10/24/2016) | Rmb64.44     |

|                                           |                |            |
|-------------------------------------------|----------------|------------|
| Visual Photonics Epitaxy Co Ltd (2455.TW) | E (09/11/2023) | NT\$358.00 |
|-------------------------------------------|----------------|------------|

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